Betting Against the Benchmark: Drift-Robust, Anytime-Valid Certificates of Out-of-Sample Predictive Ability

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Abstract

Tests of out-of-sample return predictability treat the asset's drift as a nuisance and dispose of it by estimating it. That is where they break. A forecasting model with an unpenalised intercept does not reduce to the zero forecast when its slopes vanish, so racing it against zero tests no drift *and* no conditional predictability jointly; we show in closed form that the resulting Clark–West statistic converges to the t-statistic of the asset's mean return, giving an expected value $\sqrt{n}S^2/\sqrt{S^2+(1+p)/k}$ for per-period Sharpe ratio S, estimation window k and p estimated coefficients. At Bitcoin's realised drift a model containing *no features at all* is declared significantly predictive in about two of every five samples. Replacing zero with the recursive mean removes the leading term but leaves the benchmark's own estimation error inside the statistic. We therefore eliminate the drift rather than estimate it. The *certificate* is a non-negative wealth process: one bettor stakes on the signal leading the outcome, a second hedges every candidate drift two-sidedly, and the infimum over the nuisance of their average is bounded by a martingale under the whole composite null. Validity is finite-sample and uniform over time—no bandwidth, no long-run variance, no bootstrap, no asymptotics, no refitting—and holds for arbitrary black-box pipelines, so the process can be monitored daily on a live strategy without any correction for looking. Its logarithm is the profit and loss of an explicit drift-hedged strategy, so statistical and economic significance become one number: rejecting at 5% means that strategy multiplied capital twentyfold. E-values average under arbitrary dependence, which corrects an eighteen-setting research grid without a joint bootstrap, and averaging the h phase certificates of an overlapping-horizon forecast is literally the staggered h-vintage portfolio. The construction yields a design law: certifying an annualised information ratio IR takes $2\ln(1/\alpha)/\mathrm{IR}^2$ years, about $6/\mathrm{IR}^2$ at 5%. That bound is optimal to first order, costs only three per cent more data than a fixed-sample test that must be sized in advance and looked at once, rises to $7.4/\mathrm{IR}^2$ once the drift hedge is paid for—exactly $\ln 2$ nats—and doubles again if the stake is learned online rather than declared in advance, which turns pre-registration into a power calculation. Applied to three cryptocurrencies, two horizons and three estimators over six years of daily out-of-sample data, no setting is certified, the grid-level e-value is 1.18 against the 20 required, and the anytime-valid ceilings put the features' incremental annualised information ratio between 0.6 and 2.6, every interval containing zero. That is the design law rather than the data: a sample this length cannot certify anything below an information ratio of about 1.6, so the literature's typical claims are not merely unproven but unprovable at the sample sizes on which they rest.

Preprint. Code, data, and the full result tables: https://github.com/ITheClixs/crypto-return-predictability

1 Introduction

A forecasting pipeline produces out-of-sample predictions of an asset's return, and the analyst wants to know one thing: do the features carry information, or has the model rediscovered the fact that the asset went up? The question sounds like the one the standard nested predictive-accuracy tests answer. It is not.

Those tests treat the unconditional drift as a nuisance parameter and dispose of it by plugging in an estimate—either zero, or the recursively estimated sample mean. Plugging in zero is worse than it looks. A regression fitted with an unpenalised intercept does not reduce to the zero forecast when its slopes are set to zero; it reduces to the training-window mean. Racing it against zero therefore tests *no drift and no conditional predictability* jointly, while the sentence such a test is used to support is about the features alone. Clark and West [1] are explicit that their null is a zero-mean martingale difference, so the statistic is behaving exactly as specified; the error is in the translation, and it is invisible in code. Section 4.8 makes the size of that error exact: to first order the zero-benchmark Clark–West statistic *is* the t-statistic of the asset's mean return, and on an asset with Bitcoin's drift a model with no features at all is declared significantly predictive in about two samples in five. Plugging in the recursive mean removes that leading term and leaves a smaller one, the covariance between the benchmark's own estimation error and the model's signal, which is what the residual size distortion at longer horizons consists of.

The response of this paper is not a better estimate of the drift. It is to stop estimating it.

The instrument. Section 5 constructs a *certificate*: a non-negative process $\mathcal{E}_t$ with $\mathcal{E}_0 = 1$ whose expectation never exceeds one under the composite null "the features add nothing, at any drift". Two bettors are run against each other for every candidate drift—one staking on the signal leading the outcome, one hedging the drift two-sidedly—and the certificate is the infimum over the nuisance of their average. Because an infimum is at most the value at the truth, and the value at the truth is a martingale, Ville's inequality gives $P(\exists t : \mathcal{E}_t \geq 1/\alpha) \leq \alpha$ in finite samples and uniformly in time. The proof is six lines and uses no asymptotics, no mixing condition, no long-run variance, no bandwidth and no bootstrap. It is indifferent to what produced the forecasts, so regularised, adaptively stopped and black-box learners are covered by the same statement that covers ordinary least squares—which matters, because the asymptotic theory the applied literature relies on was derived for none of them.

Three consequences that are not available to a t-statistic. First, the guarantee holds at every stopping time, so the process can be run forward on a live strategy and inspected daily with no correction for looking, and it supplies an anytime-valid *ceiling* on incremental value—the number that lets a desk retire a research line rather than merely fail to confirm it. Second, its logarithm is the profit and loss of an explicit drift-hedged strategy, so statistical and economic significance are the same quantity and rejecting at 5% means twentyfold growth of capital that was credited with no part of the market's return. Third, e-values average under arbitrary dependence, so an eighteen-setting research grid is corrected by an average and an h-step overlapping forecast by splitting capital across its h daily vintages—which is exactly the staggered portfolio the economic analysis needs, so the statistically valid combination and the implementable strategy coincide.

The law. Evidence is log wealth and log wealth grows at half the squared information ratio, so the sample-size question closes in form: certifying an annualised information ratio IR takes $2\ln(1/\alpha)/\mathrm{IR}^2$ years, about $6/\mathrm{IR}^2$ at the 5% level. Section 6 shows this is optimal to first order, so no cleverer test recovers the missing years; that continuous monitoring costs about three per cent more data than a fixed-sample test that must be pre-committed and looked at once; and that learning the stake online rather than declaring a target effect size in advance roughly doubles the requirement. Six years of daily data—about what any liquid cryptocurrency offers—cannot certify an information ratio below roughly 1.6.

The evidence. Section 8.2 applies all of this to a deliberately careful study: three assets, two horizons, six forecasters, purged and embargoed walk-forward validation over 2020–2026, programmatic leakage tests, transaction costs, execution delay and exposure regressions. Clark–West against zero rejects in five of eighteen settings and against the recursive mean in seven, and a joint resampling of the whole eighteen-test experiment puts $P(N \geq 7)$ at well under one per cent—so under

p-values there is something to explain. Under certificates there is not: no setting is certified, the grid-level e-value is 1.18 against the 20 required, the directional variant that replaces an invalid market-timing test certifies nothing either, and the ceilings bound the features' incremental annualised information ratio between 0.6 and 2.6, every interval containing zero. The two readings are not in conflict. The design law says a six-year sample cannot resolve ratios of this size, so the correct conclusion is that the question has not been answered rather than that it has been answered negatively—and that distinction is exactly what the ceiling, and not a p-value, is able to express.

What is new here, and what is not. The diagnosis in Section 4.8 is not a discovery and we state its provenance before reporting it. The zero-mean null is Clark and West's own statement of their test [1]; the recursively estimated mean has been the standard out-of-sample benchmark in the equity-premium literature since Welch and Goyal [2] and Campbell and Thompson [3]; Moosa and Burns [4] argue the drift-versus-no-drift choice directly; Pincheira et al. [5] document Clark–West's long-horizon size distortion and propose the Wild Clark–West replacement, which Magner and Hardy [6] apply to thirteen cryptocurrencies against both benchmarks. What is new in Section 4.8 is only the closed form for the distortion and its numerical validation.

The game-theoretic apparatus is likewise not ours. Test martingales, e-values and Ville's inequality are the subject of an established literature [7–9]; Wang and Ramdas [10] prove the e-BH result we use; and Choe and Ramdas [11] build confidence sequences for the difference in score between *two given* forecasters. What that literature does not contain, and what Sections 5–6 contribute, is the nested case: a comparison in which the benchmark is not given but estimated from the same data, whose nuisance is a drift shared by both forecasts, and whose null is therefore composite. The specific new objects are the drift-eliminating construction of (18) and its validity theorem, the concavity result that reduces the nuisance search to two evaluations per step, the identification of the phase-averaged certificate with the staggered portfolio, and the detection-horizon law with its optimality, anytime-validity and stake-learning corollaries. The instrument is released as a library so that the claim can be checked on someone else's pipeline rather than only on ours.

The four errors this study had to survive first. Claims of accurate cryptocurrency prediction are abundant and rarely survive scrutiny; the reported accuracy is usually traceable to target leakage [12, 13], preprocessing leakage, an absent or misspecified benchmark [14], or silent multiple testing [15]. Preprocessing leakage is impossible by construction here, because standardisation is a pipeline step fit inside each fold; target leakage is prevented by a test that perturbs future bars and asserts that no feature at or before t changes; multiple testing is handled by Theorem 5. Benchmarking is the item this study initially got wrong, and Appendix B records what was claimed, why it was wrong, and what replaced it.

2 Related work

Return predictability and out-of-sample evaluation. Welch and Goyal [2] showed that a long list of equity-premium predictors that perform well in sample fail to beat a simple historical mean out of sample, and Campbell and Thompson [3] formalised the out-of-sample R^2 against that benchmark which we adopt in Section 4.6. The lesson generalises: in-sample fit is close to uninformative about forecast value, and the appropriate benchmark is a naive one estimated recursively rather than with hindsight.

Predictive-accuracy tests for nested models. Clark and McCracken [16] give the encompassing tests for nested forecast comparison and their non-standard limiting distributions; Clark and West [1] specialise the argument to the martingale-difference null, where the benchmark is the no-change forecast and the null asserts a *zero-mean* martingale difference; Clark and West [17] extend it to general nested pairs and report that with standard normal critical values the actual size is close to, and usually a little below, nominal. The degeneracy that makes the limit non-standard—under the null the two models coincide, so the core statistic has zero variance at population parameters—is diagnosed in West [18]. Pincheira et al. [5] remove it by multiplying the larger model's error by an independent random variable with unit mean and small variance, giving the asymptotically normal Wild Clark–West statistic, which their simulations show is better sized at long horizons at some cost in power. Moosa and Burns [4] argue that whether the benchmark random walk carries a drift is an empirical question rather than a settled one, and that the answer changes which model wins. We use all of this rather than extend it; Section 4.9 measures how the resulting procedures behave in one concrete pipeline.

Machine learning in asset pricing. Gu et al. [19] conduct a large-scale comparison of machine-learning methods for equity returns and find measurable but small out-of-sample gains, concentrated in methods that shrink aggressively. Our model roster is chosen in that spirit: two regularised linear models and a shallow gradient booster (depth 3, learning rate 0.03, subsampling 0.8, $\lambda = 1$, early stopping), on the premise that an unregularised learner in this signal-to-noise regime fits noise. Those hyperparameters were fixed by hand before the grid was run and never tuned, which is a limitation and not a virtue: it keeps the search family small enough to enumerate, but it means no conclusion here concerns a model *family*.

Backtest overfitting. Bailey et al. [20] and Arnott et al. [21] document how repeated backtesting manufactures apparently significant strategies, and Bailey and López de Prado [22] propose the deflated Sharpe ratio, which raises the bar in proportion to the number of trials. White [23] and Sullivan et al. [24] address the same problem for technical trading rules through a bootstrap reality check. We adopt the deflated Sharpe ratio together with explicit control of the false discovery rate and the family-wise error rate over our grid.

Cryptocurrency market efficiency. The empirical record is mixed and appears to be time-varying. Urquhart [25] reports evidence of inefficiency in early Bitcoin returns; Nadarajah and Chu [26] find that a simple power transformation of the same returns satisfies efficiency tests, and Tiwari et al. [27] report that Bitcoin is largely efficient over most of their sample. Closest to our setting, Magner and Hardy [6] test 13 cryptocurrencies over 2018–2022 with autoregressive models and lagged Bitcoin returns, evaluated by Wild Clark–West against both the driftless random walk (a zero forecast) and the random walk with drift (a constant forecast), and report that their models beat the benchmark in a majority of exercises; Pincheira et al. [28] propose a further out-of-sample test aimed specifically at the random-walk benchmark. Reporting both benchmarks is therefore established practice in this literature, which is one reason we treat the benchmark point below as a measurement rather than a discovery. Much of this work evaluates statistical efficiency without asking whether any detected dependence survives transaction costs, which we do take up.

Game-theoretic statistics. The instrument of Section 5 is built from an established apparatus and it is worth being precise about which parts are borrowed. Test martingales, e-values and Ville's inequality [29] are developed in Shafer and Vovk [7] and surveyed by Ramdas et al. [8]; the betting construction and its predictable stakes follow Waudby-Smith and Ramdas [9]; Vovk and Wang [30] establish the merging rules for e-values and Wang and Ramdas [10] the e-BH procedure we use for false-discovery control under arbitrary dependence. Kelly [31] and Cover [32] supply the growth-rate arguments behind Section 6. Closest in intent, Choe and Ramdas [11] construct confidence sequences for the difference in score between *two given* forecasters, and Henzi and Ziegel [33] give sequentially valid inference on probability-forecast performance.

What that literature does not address is the case this paper needs. In a nested predictive comparison the benchmark is not given; it is estimated from the same data, its estimation error enters the statistic, and the nuisance it introduces—an unknown drift—is shared by both forecasts, so the null is composite rather than simple. That is precisely the structure Clark–West exists to handle, and it is the structure the game-theoretic literature has not taken up. Equation (18) and Theorem 2 are our answer to it, and Lemma 3, Remark 6 and Theorem 7 are consequences specific to that setting. On the econometric side, Giacomini and White [34] give an alternative route to valid nested inference through a fixed rolling window; it is a fixed-sample test, requires a non-vanishing estimation window, and does not remove the drift.

Validation for dependent data. Purged and embargoed cross-validation is due to López de Prado [35], and is the mechanism by which we prevent a training label whose realisation overlaps a test window from informing the fit.

Table 1: Evaluated out-of-sample window per setting. Counts are forecasts, not trades; the trading rule of Section 4.5 samples every h-th bar.

Asset	h (days)	OOS forecasts	First	Last
BTC	1	2185	2020-07-23	2026-07-16
BTC	7	2173	2020-07-29	2026-07-10
ETH	1	2185	2020-07-23	2026-07-16
ETH	7	2173	2020-07-29	2026-07-10
SOL	1	1720	2021-10-31	2026-07-16
SOL	7	1708	2021-11-06	2026-07-10

3 Data

We use daily adjusted OHLCV bars from Yahoo Finance for the USD pairs of BTC, ETH, and SOL. BTC and ETH span 2019-01-01 to 2026-07-18 (2,756 bars each); SOL begins 2020-04-10 (2,291 bars). Cryptocurrency markets trade every calendar day, so the bar index is contiguous with no weekend or holiday gaps. We verified this rather than assuming it, and Section 4.3 shows that the split logic remains conservative under gaps regardless.

The evaluated out-of-sample window is shorter than the data window because the first fold consumes 504 bars of training history. Table 1 gives the resulting per-setting counts.

Asset selection is not survivorship-free. BTC, ETH, and SOL were chosen in knowledge of their survival, while many contemporaneous tokens did not survive. This biases the study toward finding profitable long exposure, which strengthens rather than weakens a negative result about *predictability*, but would invalidate any cross-sectional claim. Section 10 returns to it.

4 Method

4.1 Features

Let C_t, H_t, L_t, V_t denote close, high, low, and volume at bar t. Every feature is a function of information available at or before t, and all twelve are scale-free so that a single model can pool across assets trading at very different price levels. We use cumulative log returns over $k \in \{1, 5, 10, 21\}$, realised volatility over $w \in \{10, 21\}$, Wilder's RSI over 14 bars recentred to roughly $[-1, 1]$, a price-normalised MACD histogram, two moving-average ratios, a mean normalised range, and a volume z-score:

$$r_t^{(k)} = \log \frac{C_t}{C_{t-k}}, \qquad \sigma_t^{(w)} = \mathrm{sd}\Big(r_{t-w+1}^{(1)}, \ldots, r_t^{(1)}\Big), \tag{1}$$

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \bar{U}_t/\bar{D}_t}, \qquad \mathrm{macd}_t = \frac{M_t - S_t}{C_t}, \tag{2}$$

$$\mathrm{ma}_t = \log \frac{\mathrm{SMA}_7(C)_t}{\mathrm{SMA}_{21}(C)_t}, \qquad \mathrm{dist}_t = \log \frac{C_t}{\mathrm{SMA}_{50}(C)_t}, \tag{3}$$

$$\mathrm{range}_t = \frac{1}{14} \sum_{i=0}^{13} \frac{H_{t-i} - L_{t-i}}{C_{t-i}}, \qquad z_t = \frac{\log(1 + V_t) - \mu_t^{(21)}}{s_t^{(21)}}, \tag{4}$$

where $\bar{U}$ and $\bar{D}$ are exponentially smoothed gains and losses with $\alpha = 1/14$, and $M_t = \mathrm{EMA}_{12}(C)_t - \mathrm{EMA}_{26}(C)_t$ with $S_t = \mathrm{EMA}_9(M)_t$. The longest warm-up is 50 bars; rows with any undefined feature are dropped.

The absence of look-ahead is enforced by a test rather than by inspection: future bars are perturbed and every feature value at or before t is asserted unchanged. This catches an accidental centred window, a sign error in a shift, or a non-causal smoother, none of which is reliably visible when reading code.

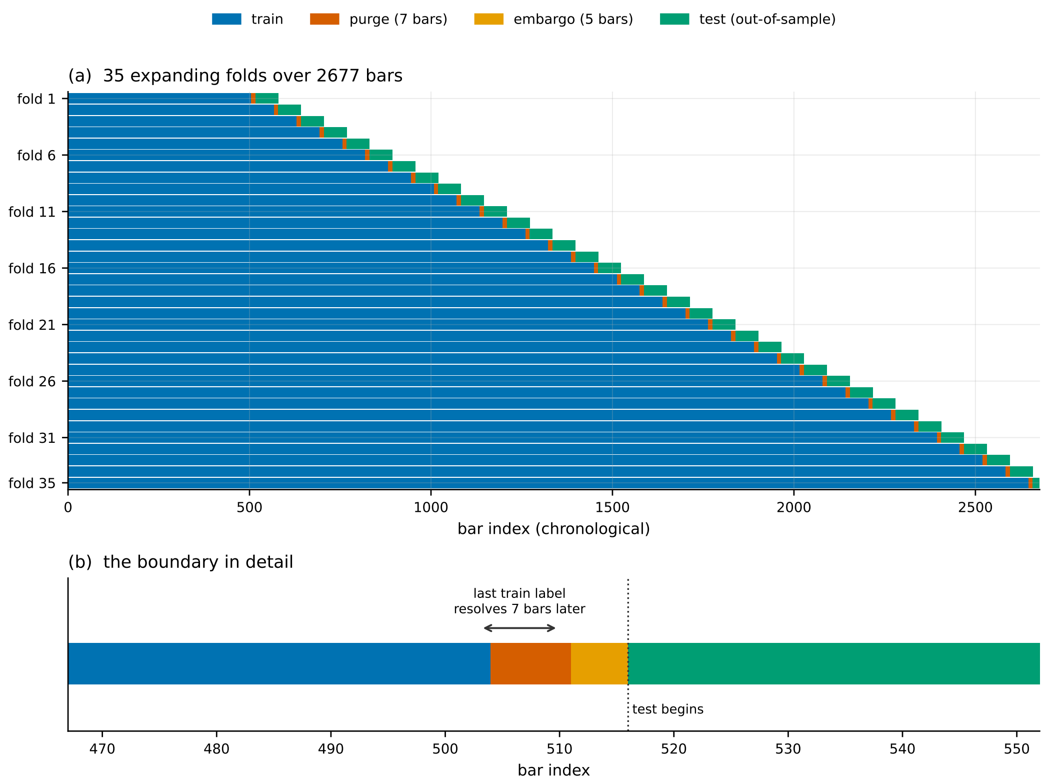

Figure 1: The validation design. Panel (a) shows all 35 folds on one timeline; panel (b) zooms on a single train/test boundary, where the purge and the embargo are individually visible. Equation (6) is what the gap enforces.

4.2 Prediction target

The label attached to a decision made at t is the return realised over $(t, t+h]$:

$$y_t^{(h)} = \log \frac{C_{t+h}}{C_t}. \tag{5}$$

The final h rows have no label. We retain them with features and a missing target, because that is exactly the state a live forecaster occupies on the most recent bar, and we exclude them from fitting and scoring.

4.3 Purged, embargoed walk-forward validation

Ordinary k-fold cross-validation shuffles time and trains on the future. A plain chronological split is also insufficient: because $y_t^{(h)}$ resolves at $t+h$, a training row close to the test block carries a label whose realisation overlaps the test window. Following López de Prado [35], we purge the h rows before each test block and impose a further embargo of e bars. Writing $\mathcal{T}_k$ and $\mathcal{V}_k$ for the train and test index sets of fold k, every fold satisfies

$$\max(\mathcal{T}_k) + h + e < \min(\mathcal{V}_k). \tag{6}$$

Test blocks tile forward without overlap. We use an expanding window with $|\mathcal{T}| \geq 504$ bars, $|\mathcal{V}| = 63$ bars, and $e = 5$, giving 35 folds for BTC and ETH and 28 for SOL. Figure 1 shows the resulting layout, drawn from the splits the study actually runs.

Purging is applied by row position, whereas the leak it prevents is a calendar one. Position-based purging remains valid under missing bars, and conservatively so: gaps only stretch the timeline, so h positions always span at least h calendar days. We assert this on a synthetic index with a quarter of its bars removed.

Two further controls are worth naming because they are commonly missed. Standardisation is a pipeline step fit inside each fold, so it never sees test rows. And the gradient booster purges its own early-stopping holdout: without a gap, the last h rows of its fitting slice carry labels realised inside that holdout, so early stopping would be tuned on partly observed outcomes. We apply the same h-bar purge one level down, and test it by corrupting exactly those rows and asserting the fitted model is unchanged.

4.4 Forecasters

Three benchmarks and three machine-learning models are refit from scratch on every fold. The benchmarks are the random walk $\hat{y}_t = 0$, the training-window mean $\hat{y}_t = \bar{y}_{\mathcal{T}}$, and an AR(1) regression of $y^{(h)}$ on $r^{(1)}$. The machine-learning models are ridge ($\alpha = 1$), elastic net ($\alpha = 10^{-3}$, $\rho = 0.5$), and XGBoost [36] with depth 3, learning rate 0.03, subsampling 0.8, $\lambda = 1$, and early stopping on the purged temporal holdout described above. The linear objectives are

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \|y - X\beta\|_2^2 + \alpha\|\beta\|_2^2, \qquad \hat{\beta}_{\text{EN}} = \arg\min_{\beta} \frac{1}{2n}\|y - X\beta\|_2^2 + \alpha\rho\|\beta\|_1 + \frac{\alpha(1-\rho)}{2}\|\beta\|_2^2. \tag{7}$$

Both linear objectives are fitted with an unpenalised intercept on standardised features, and the gradient booster estimates its own base score from the training labels. Setting the slopes to zero therefore returns the training-window mean of the target, not zero: measured on BTC at $h = 1$, the ridge and elastic-net intercepts equal $\bar{y}_{\mathcal{T}}$ to machine precision. The forecaster these models nest is the training-window mean, and the zero-return benchmark is the further restriction that the intercept is also zero. Section 4.8 shows what that distinction does to the test.

4.5 Trading rule and transaction costs

Forecasts become unit positions on a non-overlapping schedule: one decision every h bars, held to the next. Sampling every h-th bar prevents the same h-day return being counted h times. Costs are charged on turnover, so reversing a position pays for two sides:

$$w_t = \operatorname{sign}(\hat{y}_t) \in \{-1, 0, +1\}, \qquad \tau_t = |w_t - w_{t-1}|, \qquad \pi_t = w_t\left(e^{y_t^{(h)}} - 1\right) - \tau_t c, \tag{8}$$

with $c = 17$ basis points per side (10 bp taker fee, 5 bp slippage, 2 bp half-spread). Equity compounds as $E_T = \prod_{t \le T}(1 + \pi_t)$ from unity.

Because decisions are taken every h bars, the Sharpe ratio annualises at $365/h$ rather than at 365. Using the latter inflates a seven-day strategy's Sharpe by a factor of $\sqrt{7} \approx 2.6$.

The schedule admits h start offsets and nothing distinguishes them. We therefore run every strategy on all h and report the spread; at $h = 7$ it reaches 1.40 Sharpe points on one setting (Section 8.8).

An always-long reference is computed on the same schedule with the same cost model, so that a forecaster whose only achievement is holding the asset can be recognised as such.

4.6 Out-of-sample R^2_{OS}

We report the Campbell–Thompson form [3], whose denominator is the squared error of a genuine ex-ante benchmark forecast rather than of the realised mean of the evaluation window:

$$R^2_{OS} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \hat{y}_t^b)^2}, \tag{9}$$

with $\hat{y}^b$ the recursively estimated historical mean. That benchmark scores exactly zero against itself, which serves as a wiring check on the metric. The alternative denominator $\sum_t (y_t - \bar{y})^2$ uses a quantity unknowable in advance and flatters the benchmark; we compute it as a cross-check but do not quote it.

Note that this is a *different* benchmark from the one the predictive-accuracy tests of Sections 4.7 and 4.8 use, and reporting effect sizes against one reference while testing significance against another is not defensible. It also happens to be the easier reference here: the zero forecast has lower squared error than the recursive mean in all six asset–horizon cells, by R^2_{OS} of $+0.0005$ (BTC h=1) up to

$+0.1023$ (SOL h=7), the recursive drift estimate on SOL being badly wrong after the early-sample run-up. Measured against the zero forecast instead, exactly one of 18 settings has positive R^2_{OS}: BTC elastic net at $h = 1$, at $+0.0026$. Table 7 quotes the recursive-mean column; the zero-forecast comparison is the stricter one and is the number to read alongside a p-value computed against the zero forecast.

4.7 Diebold–Mariano and the nested-model problem

For loss differential $d_t = L(e_t^m) - L(e_t^b)$ under squared-error loss, the Diebold–Mariano statistic [37] is $\mathrm{DM} = \bar{d}/\sqrt{\hat{V}/n}$ with $\hat{V}$ a Newey–West long-run variance [38] truncated at $h-1$ Bartlett lags, appropriate because h-step forecast errors follow an MA($h-1$):

$$\hat{V} = \hat{\gamma}_0 + 2\sum_{k=1}^{h-1}\left(1 - \frac{k}{h}\right)\hat{\gamma}_k, \qquad \hat{\gamma}_k = \frac{1}{n}\sum_{t=k+1}^{n}(d_t - \bar{d})(d_{t-k} - \bar{d}). \tag{10}$$

The $1/n$ normalisation applies to every autocovariance including the lagged ones. This is deliberate: the Bartlett weights guarantee $\hat{V} \geq 0$ only under that normalisation, and the more intuitive $1/(n-k)$ can drive a weighted sum negative in small samples. We apply the Harvey–Leybourne–Newbold small-sample correction [39], $\mathrm{DM}^* = \mathrm{DM}\sqrt{(n+1-2h+h(h-1)/n)/n}$, against a t_{n-1} reference. A negative statistic favours the model, and the test is two-sided because the null is equal accuracy and a model can be significantly worse.

The MA($h-1$) argument justifies the bandwidth for the *forecast error* under the null. It does not carry over to $\hat{f}_t$ once Equation (12) applies, because the drift component $2\mathbb{E}[y](\hat{y}_t^m - \hat{y}_t^b)$ inherits the persistence of the forecast. At $h = 1$ the bandwidth is zero lags, so no autocovariance enters at all, and four of the five settings that reject in Section 8.2 are at $h = 1$. We therefore report both $h-1$ and the Newey–West plug-in bandwidth $\lfloor 4(n/100)^{2/9}\rfloor$, which is 8 lags at $n \approx 2{,}200$. On this data the plug-in bandwidth yields *smaller* p-values—the adjusted differentials are mildly negatively autocorrelated, so the $h-1$ choice is the conservative one at $h = 1$—but the count of surviving settings changes with it. The two bandwidths turn out to be interchangeable as far as size goes: under the independent null of Section 4.9 they differ by at most 0.3 percentage points in every cell we measured. What separates them is the rejection count on the real data, and nothing in the data selects between them, so we report the sensitivity as a result rather than choosing.

4.8 Clark–West

Under a nested null, Diebold–Mariano is undersized as a test of predictability: the larger model pays sample mean squared prediction error to estimate coefficients that are truly zero, and the test reads that cost as evidence for the benchmark. Clark and West [17] subtract the term explicitly. With $\hat{y}^b$ the nested benchmark and $\hat{y}^m$ the larger model,

$$\hat{f}_t = \left(y_t - \hat{y}_t^b\right)^2 - \left[\left(y_t - \hat{y}_t^m\right)^2 - \left(\hat{y}_t^b - \hat{y}_t^m\right)^2\right] = 2\left(y_t - \hat{y}_t^b\right)\left(\hat{y}_t^m - \hat{y}_t^b\right), \tag{11}$$

and $\bar{f}/\sqrt{\hat{V}_f/n}$ is referred to a standard normal, one-sided. A *positive* statistic favours the model, the opposite of the Diebold–Mariano convention; we report both statistics with their p-values throughout so the sign is never inferred from significance alone.

What the benchmark choice does to the hypothesis. The right-hand identity in Equation (11) is the whole issue. Clark–West is a t-test of whether the model's *deviation from the benchmark* covaries positively with the outcome's deviation from it. With $\hat{y}^b = 0$ the identity collapses to $\hat{f}_t = 2y_t\hat{y}_t^m$, whose expectation is

$$\mathbb{E}\left[\hat{f}_t\right] = 2\left(\mathbb{E}[y]\,\mathbb{E}[\hat{y}^m] + \mathrm{Cov}(y, \hat{y}^m)\right). \tag{12}$$

Only the second term is predictability. The first is drift multiplied by the average forecast, and it is nonzero whenever the sample has a drift and the model is on average long—which, through the fitted intercept, it always is here. BTC's realised drift over this sample is $+0.00102$ per day,

Table 2: Rejection rate on data containing *no* conditional predictability, as the asset's drift grows. Returns are i.i.d. with per-period standard deviation 3%; the larger model is the recursive mean plus $p = 12$ exogenous features whose coefficients are truly zero, on this study's geometry ($n = 2186$, $k = 504$). Nominal level 5%, 400 replications, Monte Carlo standard error at most 2.5 points. Clark–West against zero degrades monotonically in the drift and is reading the asset's Sharpe ratio, exactly as Proposition 1 predicts; against the recursive mean it is correctly sized in this exogenous-feature case; the certificate is bounded by its nominal level uniformly in the drift, which is Theorem 2.

	CW statistic vs. zero		rejection rate at 5%		
Annualised Sharpe	predicted	measured	vs. zero	vs. recursive mean	certificate
0.0	0.00	0.04	5.2%	5.8%	0.3%
0.4	0.42	0.17	9.0%	6.2%	0.3%
0.8	1.34	0.42	11.8%	5.8%	0.3%
1.2	2.40	1.07	33.0%	6.2%	0.3%
1.6	3.46	1.82	56.0%	5.5%	1.0%
2.0	4.50	2.60	81.0%	4.0%	0.8%
Same design with $p = 0$ (a model with no features at all):					
0.8	1.34	1.35	43.0%	—	0.0%
1.6	3.46	3.70	97.5%	—	0.0%

37% annualised. Measured across the 18 settings, that term accounts for 9% of the numerator at BTC $h = 1$ but 33% for the booster at $h = 1$ and 41–61% across BTC $h = 7$. The benchmark forecasters make the point most directly: the training-window mean, which uses no features at all, scores $\mathrm{CW} = +1.03$ against the zero forecast on BTC at $h = 1$.

How large the contamination is. Equation (12) says the drift term is present; the following says how big it gets, and it is bigger than the qualitative statement suggests.

Proposition 1 (The zero-benchmark statistic is a drift test)**.** *Let y_t be i.i.d. with mean μ and variance σ^2. Let the larger model be the intercept-only forecast $\hat\mu_t$ estimated on a window of k observations disjoint from t, plus the out-of-sample contribution of p exogenous features whose coefficients are truly zero, so that $\hat y_t^m = \hat\mu_t + u_t$ with $\mathbb{E}[u_t] = 0$ and $\mathrm{Var}(u_t) \approx p\sigma^2/k$. The correct answer is always "no incremental predictability". Then the Clark–West adjusted differential against the zero benchmark, $\hat f_t = 2y_t\hat y_t^m$, has*

$$\mathbb{E}[\hat f_t] = 2\mu^2, \qquad \mathrm{Var}(\hat f_t) \approx 4\Big(\mu^2\sigma^2 + \sigma^4\tfrac{1+p}{k}\Big),$$

so the statistic computed on n test points satisfies

$$\mathbb{E}\big[\mathrm{CW}_0\big] \;\approx\; \frac{\sqrt{n}\,S^2}{\sqrt{S^2 + (1+p)/k}}, \qquad S = \mu/\sigma.$$

For any $\mu \neq 0$ this diverges in n, so the rejection probability tends to one. Two limits are worth naming. When $kS^2 \gg 1 + p$ it reduces to $\sqrt{n}\,S$: the statistic becomes the t-statistic of the asset's mean return. *When $kS^2 \ll 1 + p$ it is $\sqrt{nk/(1+p)}\,S^2$, so a model that estimates more parameters dilutes the drift signal with its own estimation noise and the distortion shrinks—which is why the study's regularised twelve-feature models show a smaller distortion than a featureless one.*

Proof. $\hat y_t^m$ is independent of y_t with mean μ and variance $\sigma^2(1 + p)/k$, so $\mathbb{E}[2y_t\hat y_t^m] = 2\mu^2$ and $\mathrm{Var}(2y_t\hat y_t^m) = 4[\mathrm{Var}(y)\mathbb{E}[\hat y^m]^2 + \mathrm{Var}(\hat y^m)\mathbb{E}[y]^2 + \mathrm{Var}(y)\,\mathrm{Var}(\hat y^m)]$. Substituting and dividing by $\sqrt{\mathrm{Var}(\hat f)/n}$ gives the display after dropping the term of order μ^2/k. □ □

The practical content is Table 2. With this study's geometry ($n = 2186$, $k = 504$) and Bitcoin's realised drift over the sample—an annualised Sharpe ratio of 0.84—Proposition 1 predicts a mean statistic of 1.34 for a model containing *no features at all*; the measurement is 1.35, and such a model is declared significantly predictive in 43% of samples at a nominal 5%. Adding twelve estimated coefficients that are truly zero dilutes the effect to 11.8%, which is the regime this study's models occupy and is consistent with the 14–21% measured by the full-refit experiment of Section 4.9.

Either way, nothing about the features is being measured: the test is reading the asset's drift, and the applied reader is told the features predict returns. The rate rises to 81% at an annualised Sharpe ratio of 2.0, which several cryptocurrencies have printed over multi-year windows.

None of this is a fault in the statistic. Clark and West [1] define the null they test as a zero-mean martingale difference, so a drift term entering Equation (12) is the test doing what it says. The fault is in the translation: an applied paper that races a feature model against zero and reports "the features predict returns" has silently added "and the mean return is zero" to its null. We therefore report Clark–West against both the zero forecast, because that is what much of the applied literature does, and against the recursively estimated mean, which is the actual slopes-zero restriction of these estimators and the benchmark already used for R^2_{OS} in Section 4.6. Reporting one without the other conflates two hypotheses. Magner and Hardy [6] report both for cryptocurrencies, and Moosa and Burns [4] argue the general case.

4.9 Calibrating the test instead of assuming it

Whether a rejection count is surprising depends on the test's size, which we measure rather than assume. Resampling the daily log-return series into a synthetic price path makes future returns independent of every past-information feature by construction, so the null "the features add no predictive value" is true, while the realised drift, the fat tails, the feature collinearity and persistence, the sample length, the purged walk-forward geometry, and the estimators themselves are all retained. Forecasts are genuinely estimated at every origin, so the estimation-noise term Clark–West exists to remove is present—which is what distinguishes this from a simulation in which the larger model's forecast is exogenous noise. In that latter construction the population mean squared prediction errors are σ_y^2 and $\sigma_y^2 + \sigma_u^2$, so the null of equal accuracy is false rather than true, no coefficient is estimated, and Clark–West centres for the algebraic reason that $\mathbb{E}[2y_t u_t] = 0$ for any independent u however poor a forecast it is. Such a design measures neither size nor bias, and an earlier version of this paper used it.

Resampling admits three nulls, and the distinction between them turns out to matter more than any other choice in this section.

Independent. Returns are drawn iid. All dependence is destroyed, including the volatility clustering that daily crypto returns certainly have, so this measures size against a null cleaner than reality.

Block. Returns are drawn in 21-day blocks. Volatility clustering survives, but so does whatever genuine within-block predictability the real series contains, so the rejection rate is an *upper bound* on size rather than a measurement of it.

Sign-flipped. Blocks are drawn as above, then the sign of each deviation from the sample mean is randomised: $r_t = \mu + s_t(r_t^* - \mu)$ with $s_t = \pm 1$ independent. Because $|r_t - \mu|$ is untouched bar for bar, the volatility dynamics survive intact; because s_t is independent noise, $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mu$ exactly, so no conditional mean predictability remains. This is the null the other two cannot isolate: realistic dependence together with a true no-predictability hypothesis.

Table 3 gives rejection rates at a nominal 5% over 2,000 replications per cell (600 for the booster), with Monte Carlo standard errors.

Four readings follow.

First, the configuration in standard use rejects three to six times as often as nominal, and the drift-only row locates the cause: a forecaster with no feature information at all, which merely predicts the training-window mean, is called significantly better than the zero forecast in 27–42% of samples that contain no predictability whatever.

Second, substituting the recursive mean repairs it. Size is 3.1–3.7% at $h = 1$ and 5.7–5.9% at $h = 7$ under the independent null—slightly conservative, consistent with Clark and West [17]'s own report that the normal approximation is a little undersized—and it does so for the early-stopped booster as readily as for ridge. Greedy split selection, subsampling and a data-dependent number of trees do not visibly break the approximation here. That is calibration evidence for this estimator under this null, not a theorem, and we found no prior measurement of it.

Table 3: Rejection rate at a nominal 5% under three resampling nulls, with Monte Carlo standard errors. 2,000 replications per cell; 600 for the booster, which is measured at $h = 1$ only. Return pool: BTC. Sign-flipped is the null that has both realistic dependence and no conditional predictability; the block column is an upper bound on size, not a measurement. Wild Clark–West was computed on every replication and its rejection rate differs from Clark–West's by at most 0.4 percentage points in every cell, so it is not tabulated separately.

		$h = 1$			$h = 7$		
Statistic	Model	indep.	sign-flip	block	indep.	sign-flip	block
CW vs. zero forecast	ridge	12.0	14.1	53.2	22.1	28.6	25.7
CW vs. zero forecast	elastic net	17.5	20.9	43.9	23.4	30.4	25.1
CW vs. zero forecast	GBM	16.8	20.0	—	—	—	—
CW vs. recursive mean	ridge	**3.7**	**4.8**	44.6	**5.9**	**7.2**	10.7
CW vs. recursive mean	elastic net	**3.1**	**3.5**	30.6	**5.7**	**6.8**	9.5
CW vs. recursive mean	GBM	**3.7**	**3.0**	—	—	—	—
CW vs. zero, drift-only model	—	27.5	34.4	25.2	36.8	41.9	33.3
DM vs. zero, two-sided	ridge	61.0	55.5	15.0	50.6	46.9	46.8

Monte Carlo standard errors: ≤ 0.6 pp for entries below 10%, ≤ 1.1 pp elsewhere, and ≤ 1.6 pp in the 600-replication booster cells.

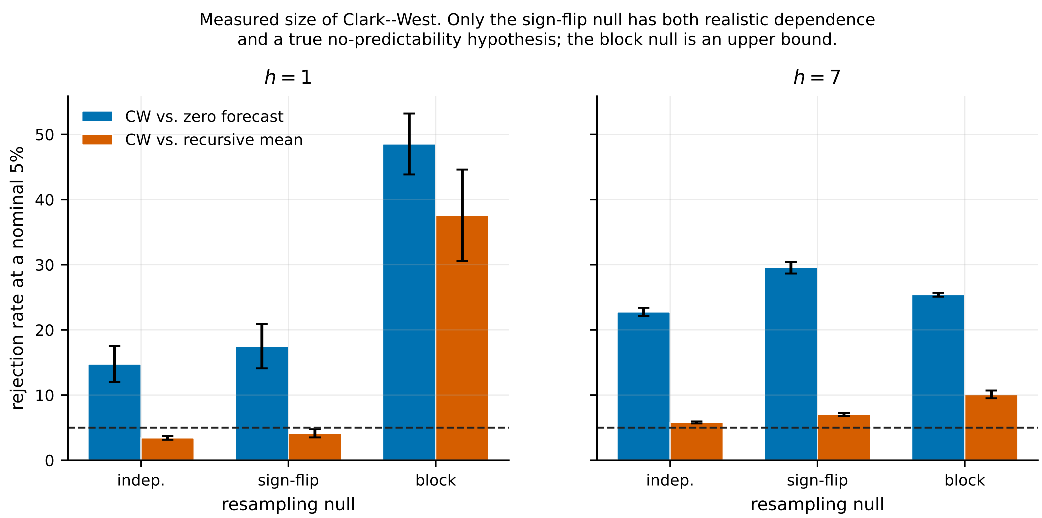

Figure 2: Measured rejection rate at a nominal 5% under the three resampling nulls. Bar height averages the estimators measured in each cell and the whisker spans them. Only the sign-flip null has both realistic dependence and a true no-predictability hypothesis; the block null is an upper bound on size, and the gap between the two is genuine within-block predictability rather than distortion in the test.

Third, the dependence question has a cleaner answer than the block column suggests. Under block resampling the corrected test rejects 30–45% of the time at $h = 1$, which the earlier version of this paper read as a size failure under dependence. It is not: under the sign-flipped null, which preserves the same volatility dynamics but removes conditional predictability by construction, size is 3.0–4.8% at $h = 1$ and 6.8–7.2% at $h = 7$. Almost all of the block column is therefore genuine within-block predictability in the real return series that block resampling carries into the synthetic path, not distortion in the test. The corrected test is usable under realistic dependence, mildly oversized at $h = 7$.

Fourth, Wild Clark–West makes no difference. Its rejection rate matches Clark–West's to within 0.4 percentage points in every cell above, including the $h = 7$ cells it was designed for. Section 8.2 gives the mechanism.

What effect would this design detect? Size answers half the question; a study that reports a null owes the other half. We index the alternative by an effect size rather than a nuisance scale: the sign-flipped deviations are passed through an AR(1) filter, $r_t - \mu = \rho\,(r_{t-1} - \mu) + \sqrt{1-\rho^2}\,d_t$, which holds the unconditional variance fixed and gives the optimal one-step predictor a population R^2_{OS} of exactly ρ^2. The one-bar return is in the feature set, so the estimators can find it.

ρ	0.02	0.04	0.06	0.08	0.10	0.11	0.12
population R^2_{OS}	0.0004	0.0016	0.0036	0.0064	0.0100	0.0121	0.0144
Power, ridge	5.2%	11.4%	24.0%	42.8%	66.2%	75.0%	83.6%
Power, elastic net	4.8%	14.0%	31.4%	55.2%	75.8%	83.8%	90.4%

Interpolating the ridge curve, the minimum detectable effect at 80% power is $\rho = 0.116$, a population R^2_{OS} of 1.34%. That is a large effect by the standards of this literature and roughly four times the largest R^2_{OS} this study observes. Two things follow, and they cut in opposite directions. Our failure to find predictability is not informative about effects below about one percent of return variance, so we make no claim there. And an effect small enough to hide from this design is small enough that the 17 bp cost model above would consume it.

Does the calibration transfer across assets? Size measured on one return distribution need not hold for another, and the rejections in Section 8.2 are spread across all three assets. Repeating the sign-flipped experiment with each asset's own return series as the pool, at 2,000 replications:

	BTC		ETH		SOL	
	h=1	h=7	h=1	h=7	h=1	h=7
CW vs. zero forecast	14.1–20.9	28.6–30.4	9.8–12.1	19.8–20.4	9.7–11.9	22.9–23.9
CW vs. recursive mean	**3.5–4.8**	**6.8–7.2**	**3.7–4.5**	**6.9–7.5**	**3.2–3.8**	**7.3–7.5**

The recursive-mean calibration transfers: 3.2–4.8% at $h = 1$ and 6.8–7.5% at $h = 7$ across three assets whose daily volatilities differ by a factor of two. The zero-benchmark distortion does not transfer, and Equation (12) says it should not—it scales with drift relative to noise, so BTC, with the smallest volatility and a substantial drift, is the worst affected.

4.10 Sign timing

A model can be poor at magnitude and useful at direction, which is all a sign strategy requires. Pesaran and Timmermann [40] test independence between the sign of the forecast and the sign of the outcome; their statistic S is asymptotically standard normal, positive for sign-timing skill and negative for a reliably wrong-way forecast. Appendix C states it. The assumption that matters here is that every variance term in it divides by n, so the n observations must be independent. Consecutive h-step forecasts built from overlapping windows are not, and Section 8.6 reports what that did to our own numbers. The statistic is undefined when the forecast never changes sign, which is the case for the always-long drift models, and we report it as missing rather than as a number.

4.11 Sharpe ratio, deflation, and interval estimates

Sharpe ratios are noisy in short samples [41] and inflated by selection. We report the probabilistic Sharpe ratio of Bailey and López de Prado [22], the probability that the true Sharpe exceeds a threshold SR^* after correcting for sample length, skewness and kurtosis, and the deflated Sharpe ratio, which sets SR^* to the expected maximum over N trials so that the bar rises with the size of the search. Appendix C states both. Here N is the number of models raced within one asset–horizon, which is a floor on the true search and not an estimate of it, so the reported deflated Sharpe is an upper bound and is labelled as such. Confidence intervals come from a circular block bootstrap [42] with block length $\lfloor n^{1/3} \rceil$ and 500 resamples, reported as percentile intervals.

4.12 Multiple testing

Eighteen settings tested by a correctly sized 5% test would produce $0.05 \times 18 = 0.9$ rejections from noise alone. That is the right calculation only for a test with the size it claims. Averaging the sign-flipped column of Table 3 over the measured model–horizon combinations, the expected count is about 4 against the zero forecast and about 1 against the recursive mean. The zero-benchmark expectation is more than four times the nominal one, and the recursive-mean expectation is close to it. These, not 0.9, are the numbers a raw count must be read against, and which of them applies depends entirely on the benchmark. We report raw counts alongside two adjustments of the Clark–West p-values over the family of 18. Holm–Bonferroni [43] controls the family-wise error rate and Benjamini–Hochberg [44] the false discovery rate:

$$p_{(i)}^{\mathrm{Holm}} = \max_{j \le i} \min\bigl\{(m-j+1)\, p_{(j)},\, 1\bigr\}, \qquad p_{(i)}^{\mathrm{BH}} = \min_{j \ge i} \min\Bigl\{\tfrac{m}{j}\, p_{(j)},\, 1\Bigr\}. \tag{13}$$

Benchmarks are excluded from the family: they are the null being tested against, not candidates in the search.

5 Certificates of incremental predictive ability

Section 4.8 located the problem in the benchmark and Section 4.9 measured what it costs. Neither repairs it. Replacing the zero forecast with the recursive mean removes the leading drift term but leaves the residual $-\mathbb{E}[(\hat{\mu}_t - \mu)\, g_t]$, the covariance between the benchmark's own estimation error and the model's signal; that residual is what the 7–8% measured size at $h = 7$ in Table 3 consists of. The deeper difficulty is structural. Every test in this family handles the drift by *estimating* it, and an estimated nuisance leaves an estimation error inside the statistic.

This section eliminates the drift instead. The construction is a wealth process, and the three properties that follow—finite-sample validity for the composite null, a growth rate that equals the strategy's own profit and loss, and a merging rule that needs no dependence assumption—are theorems rather than simulation results.

5.1 Setup

Let $(\mathcal{F}_t)_{t\ge 0}$ be a filtration, y_t the realised outcome at t and g_t any $\mathcal{F}_{t-1}$-measurable *signal*. In a nested forecast comparison the signal of interest is

$$g_t \;=\; \hat{y}_t^{\,m} - \hat{y}_t^{\,b}, \tag{14}$$

the model's forecast minus the intercept-only forecast produced by the same estimation window and refitting schedule, so that the hypothesis under test is exactly the one an applied paper means: *the features add nothing beyond the intercept*. Validity, as will be seen, does not depend on this choice; only power does.

We test the composite null

$$H_0 \;=\; \bigcup_{\mu \in \mathbb{R}} P_\mu, \tag{15}$$

where under P_μ the sequence $y_t - \mu$ is a martingale difference with respect to $(\mathcal{F}_t)$. The union over μ is the point: the drift is a nuisance and is not assumed, estimated or plugged in. Two variants of the test are available and they differ only in what they ask of P_μ.

Assumption M (martingale). $\mathbb{E}[y_t \mid \mathcal{F}_{t-1}] = \mu$ and $|y_t| \le R$ almost surely for a constant R fixed in advance.

Assumption S (symmetry). $y_t - \mu$ is conditionally symmetric about zero given $\mathcal{F}_{t-1}$.

Assumption S implies Assumption M without the envelope. It is stronger than a martingale difference and far weaker than independence: it permits arbitrary volatility clustering, arbitrary tail thickness, and any dependence whatsoever in $|y_t - \mu|$. It is also exactly the null the sign-flipped resampling of Section 4.9 constructs, which is why the two devices agree.

Fix an $\mathcal{F}_{t-1}$-measurable scale $s_t > 0$ and an odd payoff ϕ. Under Assumption S any bounded odd ϕ satisfies $\mathbb{E}[\phi((y_t - \mu)/s_t) \mid \mathcal{F}_{t-1}] = 0$, because scaling by a positive predictable quantity preserves conditional symmetry; we use $\phi = \tanh$ for magnitude and $\phi = \mathrm{sign}$ for direction. Under Assumption M only $\phi(u) = u$ is available, and the envelope R enters the stake ceiling below.

5.2 Construction

For a candidate drift μ write $\phi_t(\mu) = \phi\big((y_t - \mu)/s_t\big)$ and let $\lambda_t \geq 0$ and $\kappa_t > 0$ be $\mathcal{F}_{t-1}$-measurable stakes obeying $\lambda_t |g_t| B_t \leq c < 1$ and $\kappa_t B_t \leq c < 1$, where B_t is a predictable bound on $|\phi_t|$ ($B_t = 1$ for bounded ϕ; $B_t = (R + \max|\mu|)/s_t$ for the identity payoff). Define

$$W_t^{\text{sig}}(\mu) = \prod_{u \leq t} \Big(1 + \lambda_u\, g_u\, \phi_u(\mu)\Big), \tag{16}$$

$$W_t^{\text{ctr}}(\mu) = \tfrac{1}{2} \prod_{u \leq t} \Big(1 + \kappa_u \phi_u(\mu)\Big) + \tfrac{1}{2} \prod_{u \leq t} \Big(1 - \kappa_u \phi_u(\mu)\Big), \tag{17}$$

$$\mathcal{E}_t = \inf_{\mu \in \mathbb{R}} \ \tfrac{1}{2}\Big(W_t^{\text{sig}}(\mu) + W_t^{\text{ctr}}(\mu)\Big). \tag{18}$$

The two processes are two bettors. The first stakes on the signal genuinely leading the outcome. The second stakes, two-sidedly, on μ being the wrong centre. At the true drift the second bettor breaks even and the infimum is driven by the first; at a wrong drift the second bettor gets rich, so candidates the data have ruled out cannot drag the infimum down. The nuisance is hedged, not estimated.

Theorem 2 (Validity)**.** *Under Assumption S (or Assumption M with ϕ the identity), for every $P_{\mu_0} \in H_0$ and every $\alpha \in (0, 1)$,*

$$P_{\mu_0}\Big(\exists\, t \geq 1 :\ \mathcal{E}_t \geq 1/\alpha\Big) \ \leq\ \alpha.$$

Consequently $1/\sup_t \mathcal{E}_t$ is a p-value valid at every stopping time, and $\mathcal{E}_\tau$ is an e-value for any stopping time τ.

Proof. Fix $\mu = \mu_0$. Since ϕ is odd and $(y_u - \mu_0)/s_u$ is conditionally symmetric about zero given $\mathcal{F}_{u-1}$, $\mathbb{E}[\phi_u(\mu_0) \mid \mathcal{F}_{u-1}] = 0$; under Assumption M with $\phi(u) = u$ the same holds by the martingale property. The multipliers $\lambda_u g_u$ and $\pm\kappa_u$ are $\mathcal{F}_{u-1}$-measurable, so each factor in (16) and (17) has conditional expectation one and, by the stake ceilings, is strictly positive. Each product is therefore a non-negative martingale started at one; W^{ctr} is an average of two such, and the bracket in (18) is an average of two, hence itself a non-negative martingale started at one. Finally $\mathcal{E}_t \leq \frac{1}{2}(W_t^{\text{sig}}(\mu_0) + W_t^{\text{ctr}}(\mu_0))$ pointwise, because an infimum is at most the value at μ_0. Ville's inequality applied to that martingale gives the bound. □ □

Nothing in the proof uses asymptotics, mixing conditions, a long-run variance estimate, a bandwidth, or a bootstrap. Nothing in it is disturbed by the estimator inside the pipeline being regularised, adaptively stopped, or a black box: g_t enters only through being known one step ahead of the outcome. And nothing in it is disturbed by looking at $\mathcal{E}_t$ every day, which is what makes the process usable as a live monitor.

Lemma 3 (Concavity, and the fast path)**.** *For $\phi(u) = u$, $\mu \mapsto \log W_t^{\text{sig}}(\mu)$ is concave, so its infimum over any interval is attained at an endpoint.*

Proof. $\partial_\mu^2 \log\big(1 + \lambda_u g_u (y_u - \mu)/s_u\big) = -\big(\lambda_u g_u / s_u\big)^2 / \big(1 + \lambda_u g_u (y_u - \mu)/s_u\big)^2 \leq 0$, and a sum of concave functions is concave. □ □

Lemma 3 matters operationally rather than theoretically: with the identity payoff the drift search costs two evaluations per step instead of a grid, so the certificate runs in $O(n)$ time and $O(1)$ memory and can be updated online, one observation at a time. For bounded odd payoffs concavity can fail and the infimum in (18) is evaluated numerically; Section 7 reports the grid refinement.

5.3 The statistic is a profit and loss

Theorem 4 (Evidence in units of capital)**.** $\log \frac{1}{2}\big(W_T^{\text{sig}}(\mu) + W_T^{\text{ctr}}(\mu)\big)$ *is the cumulative log return, net of the drift, of a self-financing strategy that at each date holds $\lambda_t g_t$ units of a claim paying $\phi_t(\mu)$ and splits capital equally with a two-sided hedge of the drift. Rejecting H_0 at level α requires that strategy to have multiplied its capital by $1/\alpha$.*

The theorem is immediate from (16)–(18)—each factor is one plus a period return—but its content is not. Statistical significance and economic significance are usually separate quantities that a study reports side by side and hopes will agree; here they are the same number, and the exchange rate is fixed. At $\alpha = 0.05$ the requirement is a twentyfold gain, $\log 20 = 3.00$ nats, earned by a strategy that is credited with no part of the asset's drift. A model that is long the market and nothing else earns none of it, by construction rather than by a regression on buy-and-hold run afterwards.

This also dissolves a question the two-metric literature keeps asking. When a study's statistically significant settings and its economically profitable settings are disjoint, the natural reading is that one metric is measuring something the other is not. Under Theorem 4 there is only one metric, and disjointness cannot arise.

5.4 Multiplicity and overlap, without a bootstrap

Theorem 5 (Merging). *Let $E_1, \ldots, E_m$ be e-values for the same null, and $w_i \geq 0$ with $\sum_i w_i = 1$. Then $\sum_i w_i E_i$ is an e-value,* whatever the dependence between the E_i. *If the E_i are e-values and $\alpha \in (0, 1)$, the e-Benjamini–Hochberg procedure—reject the k largest for the largest k with $E_{(k)} \geq m/(\alpha k)$—controls the false discovery rate at α under arbitrary dependence [10].*

of the first claim. $\mathbb{E}[\sum_i w_i E_i] = \sum_i w_i \mathbb{E}[E_i] \leq \sum_i w_i = 1$ by linearity, which uses no joint distribution. □ □

This is the whole of the multiple-testing apparatus this paper needs. The eighteen settings share features, assets, horizons, market regimes and two nearly identical estimators; under p-values that dependence forces either a conservative bound or a joint resampling experiment, and Section 8.5 runs the latter for comparison at a cost of several hours of computation. Under e-values the average of the eighteen certificates is already a valid test of the global null, and no dependence needs to be modelled, estimated or simulated.

Overlapping horizons. An h-step forecast made every day produces overlapping outcomes, so the per-day payoffs are not a martingale difference sequence and no wealth process can be built on them directly. This is the defect that invalidates the averaged Pesaran–Timmermann statistic of Section 4.10: h phase-specific standardised statistics were averaged and the average referred to a standard normal, without a variance for the average. Partitioning $\{1, \ldots, n\}$ into the h phases $P_j = \{j, j + h, j + 2h, \ldots\}$ fixes it. Within a phase, consecutive outcomes span disjoint windows and the payoffs *are* a martingale difference sequence with respect to the thinned filtration, so each phase yields a certificate; Theorem 5 then merges the h of them with no assumption about how they depend on one another.

Remark 6 (The valid combination is the implementable portfolio). Averaging the h phase certificates means splitting capital $1/h$ across h bettors, one per daily vintage, and reading off their aggregate terminal wealth. That is precisely the staggered portfolio the economic section adopts in Section 8.7: an equal allocation to each of the h daily vintages, each held h days. The statistically valid combination and the economically natural implementation are the same object, so the arbitrary choice of starting phase disappears from both at once.

6 How much data an alpha needs

Because evidence is log wealth and log wealth grows at a rate fixed by the signal's information ratio, the sample-size question has a closed-form answer.

Theorem 7 (Detection horizon). *Let the signal-weighted strategy have per-period information ratio IR_p, small, and let the stake be the Kelly fraction. Then expected log wealth grows at $\mathrm{IR}_p^2/2$ nats per period, and the expected time to reach the level-α threshold is*

$$T^* = \frac{2\ln(1/\alpha)}{\mathrm{IR}_p^2} \textit{ periods} = \frac{2\ln(1/\alpha)}{\mathrm{IR}^2} \textit{ years},$$

where $\mathrm{IR} = \mathrm{IR}_p\sqrt{P}$ is the annualised ratio at P periods a year.

Table 4: The detection horizon of Theorem 7, in years of daily data at $\alpha = 0.05$ and 365 periods a year. "Pre-committed" fixes the stake at the Kelly fraction for the target ratio and pays the $\ln 2$ drift hedge; "learned online" adds the $\frac{1}{2}\ln T$ regret of Corollary 11. The fixed-sample column is a one-sided test at 80% power that must be sized in advance, looked at once, and that assumes the benchmark known. Simulated median detection times over 100 replications agree: 1.93 against 1.84 predicted at $\mathrm{IR} = 2$, 0.89 against 0.82 at $\mathrm{IR} = 3$.

Annualised IR	Pre-committed stake	Learned online	Fixed-sample
0.30	82.0	206.8	68.7
0.50	29.5	70.1	24.7
0.75	13.1	29.6	11.0
1.00	7.4	16.1	6.2
1.50	3.3	6.7	2.7
2.00	1.8	3.6	1.5
Read backwards: smallest IR *a sample of this length can certify*			
6.0 years (this study)		1.11	1.59
10 years		0.86	1.25
25 years		0.54	0.81

sketch. With $\psi_t = g_t\phi_t$, $\sup_\lambda \mathbb{E}\log(1+\lambda\psi) = \mathbb{E}[\psi]^2/(2\mathbb{E}[\psi^2])+O(\mathbb{E}[\psi]^3)$. For the identity payoff and g_t proportional to the conditional mean m_t, $\mathbb{E}[\psi] = \mathbb{E}[m^2]$ and $\mathbb{E}[\psi^2] = \mathbb{E}[m^2]\sigma^2$ to leading order, so the rate is $\mathbb{E}[m^2]/(2\sigma^2) = \mathrm{IR}_p^2/2$. Rejection needs $\ln(1/\alpha)$ nats. The full statement and the bounded-payoff case are in Appendix A. □ □

Corollary 8 (Six over IR squared)**.** *At $\alpha = 0.05$, $T^* \approx 6/\mathrm{IR}^2$ years: six years for an information ratio of 1.0, twenty-four for 0.5, sixty-seven for 0.3.*

Corollary 9 (The bound is not an artifact of this instrument)**.** *The Kullback–Leibler divergence per period between the alternative and H_0 is $\mathrm{IR}_p^2/2 + o(\mathrm{IR}_p^2)$. By the standard sequential lower bound, no level-α anytime-valid test has expected stopping time below $\ln(1/\alpha)/\mathrm{KL}$, so T^* is optimal to first order and no cleverer test recovers the missing years.*

Corollary 10 (Anytime-validity is nearly free; the drift hedge is not quite)**.** *A one-sided fixed-sample test at $\alpha = 0.05$ with 80% power needs $(z_{0.05} + z_{0.20})^2/\mathrm{IR}^2 = 6.18/\mathrm{IR}^2$ years, against $5.99/\mathrm{IR}^2$ for the sequential law: continuous monitoring costs about three per cent more data and buys the right to stop, restart, and look every day. Eliminating the drift does cost something, and the amount is exact. Averaging the signal bettor with the centring bettor in* (18) *halves any terminal wealth the signal earns, which is $\ln 2 = 0.69$ nats on a budget of $\ln(1/\alpha) = 3.00$: the certificate needs $7.38/\mathrm{IR}^2$ years, 23% more than a test that is handed the true drift. That is the whole price of never estimating it.*

Corollary 11 (Not knowing your own stake is not free)**.** *If the Kelly fraction must be learned online, log wealth falls short of the oracle's by a regret ρ_T, with $\rho_T = \frac{1}{2}\ln T + O(1)$ attainable and unimprovable in order. The horizon then solves $T\,\mathrm{IR}_p^2/2 = \ln(1/\alpha) + \rho_T$. At $T \approx 2{,}000$ daily periods $\rho_T \approx 3.8$ nats, larger than the 3.0 nats needed to reject: the cost of not knowing how strong the signal is exceeds the entire evidentiary budget, and multiplies T^* by between two and three. Declaring in advance the smallest information ratio worth detecting fixes the stake and removes the term.*

Corollary 11 converts a methodological platitude into a power calculation. A pre-registered effect size is not a formality here; it is worth a factor of two in data. Section 8.3 therefore reports both variants, with the design ratio declared before the data were touched.

Table 4 reads the law off for the ratios an applied paper might hope to find. The column that matters for this literature is the last: with the six years of daily out-of-sample data this study has—and daily crypto histories are not much longer than this for any liquid asset—nothing below an annualised information ratio of about 1.6 can be certified at all, and about 1.1 even with the stake pre-committed. Reported Sharpe ratios in the applied cryptocurrency literature are routinely in the 0.5–1.5 range and are almost never separated from market exposure. Corollary 9 says the shortfall cannot be argued away by a better test.

7 Implementation

The certificate is released as `alphacert`, a dependency-light package that consumes walk-forward forecasts a pipeline has already written and returns the wealth path, the anytime-valid p-value, the merged grid-level e-value, the e-BH selection, and the design quantities of Section 6.

Cost. Calibrating a Clark–West-family test on a machine-learning pipeline means simulating the null and refitting the entire pipeline in every replication: Section 4.9 required 2,000 refits of three assets, two horizons and three estimators, several hours on a workstation, and it must be redone whenever the feature set, the estimator, or the sample changes. The certificate needs none of it, because Theorem 2 is a finite-sample statement rather than an approximation to be checked. One pass over the existing forecasts costs $O(nG)$ for a drift grid of G points, or $O(n)$ under Lemma 3; on this study a certificate takes about 0.1–0.5 seconds against several hours for the corresponding calibration. That gap is the difference between an instrument that runs nightly on every live strategy and one that occupies a cluster for a weekend.

The one numerical trap. The infimum in (18) is over a continuum and is evaluated on a grid. The grid must resolve the drift to better than its own sampling error $\sigma/\sqrt{n}$: a coarser grid does not merely lose precision, it lifts the numerical infimum above the true one and the process stops being conservative. On this study $\sigma/\sqrt{n} = 1.7 \times 10^{-3}$; at spacings of 5×10^{-4}, 2.5×10^{-4} and 1×10^{-4} no e-value moves in the fourth decimal, and the coarsest is five times faster. The library defaults to 10^{-4}, exposes the spacing, and its test suite records the failure mode of a deliberately coarse grid so that the trap is documented rather than latent.

Live use. Because the guarantee holds at every stopping time, the wealth path can be run forward on a deployed strategy and inspected daily without any correction for looking. Two decision rules follow directly. A strategy is *certified* the first time its wealth crosses $1/\alpha$. A research line can be *retired* when the anytime-valid ceiling of Section 8.4 falls below the trading cost hurdle—a rigorous statement that the edge, if any, is too small to pay for, available at any time and without a pre-committed sample size.

8 Results

8.1 Point accuracy

Out-of-sample R^2_{OS} is negative in 16 of 18 machine-learning settings, from -0.0908 (ETH, $h = 7$, ridge) to $+0.0031$ (BTC, $h = 1$, elastic net). Rank information coefficients lie between -0.080 and $+0.053$. Figure 3 gives the full surface and Table 7 the underlying numbers.

That ridge attains CW $= +2.62$ with $p = 0.004$ on BTC at $h = 1$ while its R^2_{OS} is -0.0079 has two explanations, and only one of them was in the earlier version of this paper. Clark–West does ask about *population* mean squared prediction error, and a model can have a lower population error while its own parameter-estimation noise leaves the realised forecast worse than a constant. But the two numbers are also computed against different benchmarks—the p-value against zero, the R^2_{OS} against the recursive mean—and, per Equation (12) and Table 3, the p-value comes from a test that rejects 14% of the time when nothing is predictable. The gap is therefore not usable as evidence of a weak signal: both readings are available from the same forecasts, and the second explanation requires no signal at all.

8.2 Predictive accuracy against the random walk

Table 5 summarises. Under Diebold–Mariano the conclusion would be that machine learning is actively harmful here: no setting beats the random walk and seven lose to it significantly. Under Clark–West against the same zero benchmark, five settings reject at an uncorrected 5%. Both numbers derive from identical forecasts, and neither is the answer, because the second test rejects 14–30% of the time when nothing is predictable (Table 3).

Ordered by p-value the five are BTC ridge ($p = 0.004$), BTC elastic net (0.005), BTC GBM (0.022), and ETH ridge (0.031), all at $h = 1$, together with BTC elastic net at $h = 7$ (0.039). The sixth,

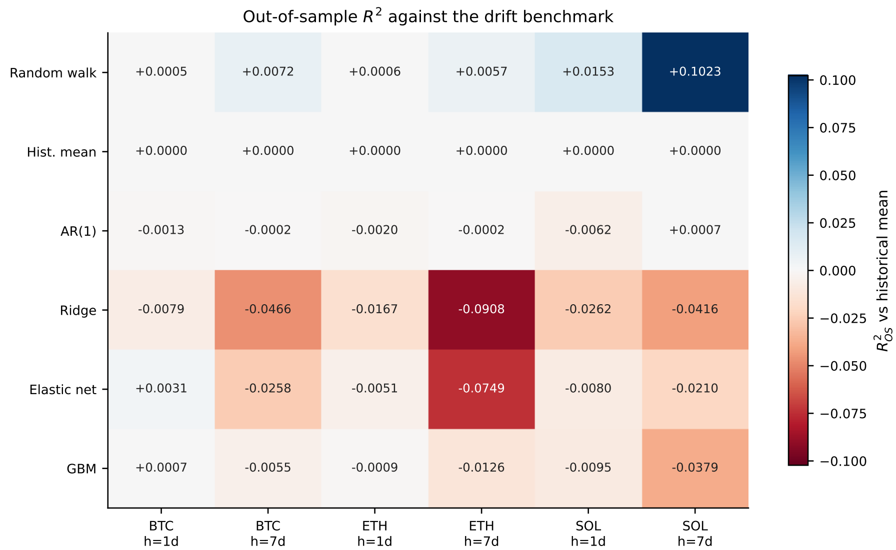

Figure 3: Out-of-sample R^2_{OS} against the drift benchmark. Red is worse than an ex-ante historical-mean forecast, blue better. The benchmark row is exactly zero by construction, which fixes the scale.

BTC ridge at $h = 7$, misses at $p = 0.05004$; a count that turns on four parts in 10^5 should not be described as five.

The benchmark changes which settings reject, not how many. Substituting the recursively estimated mean—the actual slopes-zero restriction of these estimators—moves the raw count from 5 of 18 to 7 of 18 at the same bandwidth. It does not move the same settings. BTC ridge and elastic net at $h = 1$ survive ($p = 0.004$ both). BTC at $h = 7$ drops out ($p = 0.086$ and 0.061). And SOL, which shows no signal at all against the zero benchmark, becomes the strongest rejecter: SOL GBM at $h = 1$ goes from $p = 0.207$ to 0.031, SOL ridge at $h = 7$ from 0.923 to 0.022, SOL elastic net at $h = 7$ from 0.921 to 0.011. ETH at $h = 7$ moves further against the models.

The earlier reading of this table—four of five rejections in BTC, consistent with a weak short-horizon effect in the most liquid asset—was an artifact of the benchmark. Under the corrected benchmark the geography reverses.

Calibration then does something different to each count, and this is the result we would most like a reader to take away. Against the zero forecast, 5 of 18 is close to the ≈ 4 that the sign-flipped null produces when nothing is predictable, so the zero-benchmark rejections carry essentially no information. Against the recursive mean, 7 of 18 is well above the ≈ 1 expected, and that excess is not accounted for by size. The two benchmarks therefore support opposite readings of the same forecasts: under the default one the study has nothing to report, and under the one the estimators actually nest it has a small excess to report.

We are careful about how much that excess is worth. It does not survive family-wise control at any bandwidth, and only 2 of 18 survive false-discovery control at the $h - 1$ bandwidth (5 at the plug-in bandwidth, with 0 and 1 surviving family-wise control respectively); since nothing in the data selects the bandwidth, the surviving set is not identified by this design. The two false-discovery survivors are BTC ridge and BTC elastic net at $h = 1$, which with $\alpha = 10^{-3}$ are close to the same estimator on the same data, so they are one candidate finding rather than two. Every other measurement in this paper points the other way: those settings have negative R^2_{OS} against the same benchmark, no sign-timing skill, and no alpha once market exposure and a one-bar execution delay are accounted for.

Wild Clark–West changes nothing here. Pincheira et al. [5] restore asymptotic normality by multiplying the larger model's error by $\theta_t \sim \mathcal{N}(1, \phi^2)$ with $\phi = c\,\widehat{\mathrm{sd}}(\hat{e}^{\,m})$ and c a small constant (Appendix C). Applied to these forecasts, the statistic is numerically almost identical to Clark–West:

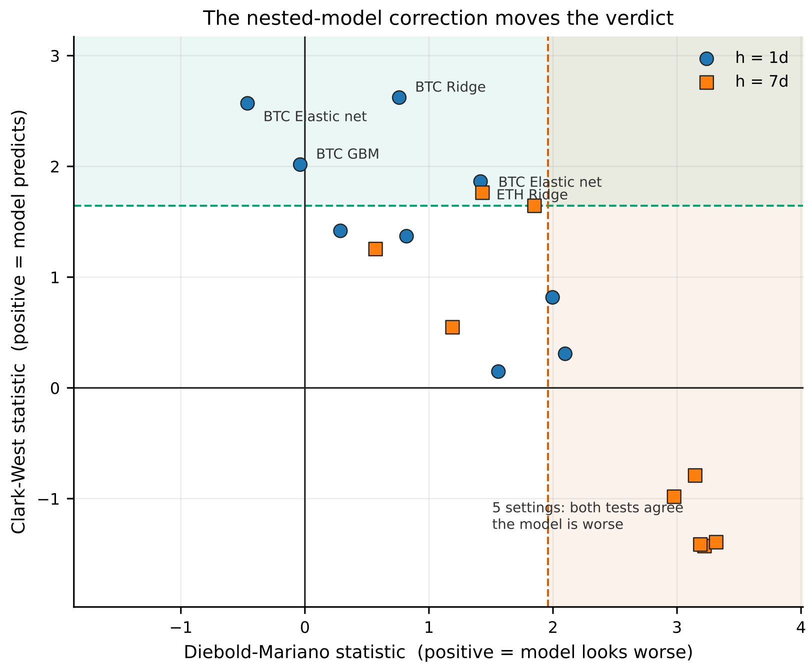

Figure 4: Every setting placed by its Diebold–Mariano statistic against its Clark–West statistic. The shaded vertical band marks where DM calls the model significantly worse; the horizontal band marks where Clark–West rejects no-predictability. Points falling in both are the settings on which the two tests give opposite readings of identical forecasts, which under nesting is expected rather than anomalous.

Table 5: Outcome of each test and correction across the 18 machine-learning settings. Direction is required, not merely significance: a Diebold–Mariano rejection is attributed to whichever side the statistic's sign favours.

Criterion	Settings
Diebold–Mariano, favours model	0 / 18
Diebold–Mariano, favours benchmark	7 / 18
Clark–West, uncorrected	5 / 18
Clark–West, Benjamini–Hochberg	2 / 18
Clark–West, Holm–Bonferroni	0 / 18
Pesaran–Timmermann, sign skill	0 / 18
Pesaran–Timmermann, anti-predictive	0 / 18
Net Sharpe interval excludes zero	2 / 18
Expected by chance at $\alpha = 0.05$	0.9 / 18

across all 36 setting–benchmark pairs the largest absolute difference is 0.0034 at $c = 0.04$ and 0.0019 at $c = 0.02$, and no rejection decision at the 5% level changes under any c in the authors' grid. The reason is a scale mismatch specific to this regime. The perturbation's standard deviation is calibrated to the forecast *error*, $\phi = c\,\widehat{\mathrm{sd}}(\hat{e}^{\,m})$, whereas the core statistic's own scale is set by the forecast *difference* $\hat{y}^{\,m} - \hat{y}^{\,b}$. Here the second is far smaller than the first, and the injected term contributes between 0.5% and 5.1% of the core statistic's standard deviation at $c = 0.02$ (median 1.5%), so the statistic barely moves. The single setting where the draw matters is BTC ridge at $h = 7$, whose p-value is 0.05004: there the 5% decision goes to the model in 43% of draws. Separately, and more fundamentally, $\mathbb{E}[\theta_t] = 1$ leaves the drift term of Equation (12) untouched, so no version of the statistic can repair a benchmark that encodes the wrong hypothesis.

Table 6: Clark–West against certificates, on the same eighteen settings and the same forecasts. The signal is the model's forecast minus the intercept-only forecast from the same window and refit schedule; at $h = 7$ the seven phase certificates are averaged (Remark 6). "Designed" pre-commits the stake to an annualised information ratio of 1.0, declared before the data were examined. Rejecting at $\alpha = 0.05$ needs an e-value of 20; the largest here is 1.81. "Implied IR" reads the realised growth rate back through Theorem 7; "ceiling" is the 95% anytime-valid upper bound of Section 8.4, and every corresponding interval contains zero. This sample can certify nothing below $\mathrm{IR} = 1.59$ (1.11 designed).

			Clark–West		e-value		information ratio	
Asset	h	Model	vs. zero	vs. mean	learned	designed	implied	ceiling
BTC	1	ridge	2.62	2.66	1.00	0.96	0.00	2.20
BTC	1	elastic_net	2.57	2.67	1.00	0.96	0.00	1.87
BTC	1	gbm	2.02	1.75	1.00	0.96	0.00	1.13
BTC	7	ridge	1.64	1.36	1.81	0.99	0.42	1.68
BTC	7	elastic_net	1.76	1.54	1.66	0.99	0.39	1.54
BTC	7	gbm	1.25	1.02	1.04	1.19	0.11	2.55
ETH	1	ridge	1.86	1.76	1.05	1.33	0.12	2.00
ETH	1	elastic_net	1.37	1.25	1.10	0.96	0.18	1.73
ETH	1	gbm	1.42	1.08	1.00	0.96	0.00	1.88
ETH	7	ridge	−0.79	−1.62	1.12	1.01	0.18	0.70
ETH	7	elastic_net	−0.98	−1.80	1.11	1.01	0.17	0.63
ETH	7	gbm	0.55	−0.36	1.03	1.01	0.09	0.78
SOL	1	ridge	0.31	1.34	1.82	0.98	0.50	2.39
SOL	1	elastic_net	0.15	1.43	1.00	1.58	0.00	2.44
SOL	1	gbm	0.82	1.87	1.20	1.22	0.28	1.78
SOL	7	ridge	−1.43	2.01	1.11	1.11	0.20	1.86
SOL	7	elastic_net	−1.41	2.28	1.11	1.11	0.20	1.95
SOL	7	gbm	−1.39	0.37	1.14	1.12	0.23	1.01
Grid-level e-value (average; Theorem 5)					1.18		20 required	
Directional variant, $\phi = \text{sign}$					1.07		20 required	
Settings surviving e-BH at 5%					0		of 18	

8.3 Certificates

Table 6 runs the certificate of Section 5 on the same eighteen settings, using as signal the model's forecast minus the intercept-only forecast from the same window and refit schedule. Nothing is refit: the certificate consumes the forecasts the pipeline already wrote. At $h = 7$ the sample is split into its seven phases and the phase e-values are averaged, per Remark 6.

Nothing is certified. The largest e-value in the grid is 1.81 against the 20 required at $\alpha = 0.05$; the average over the eighteen settings, which is by Theorem 5 a valid test of the global null under arbitrary dependence, is 1.18; e-BH selects nothing. Pre-committing the stake to an annualised information ratio of 1.0—declared before the data were touched, as Corollary 11 recommends—does not change the reading: the largest designed e-value is 1.58.

The directional variant, $\phi = \text{sign}$, is the anytime-valid replacement for the market-timing test of Section 4.10 and inherits none of its defects: it needs no asymptotic variance, and its phase average requires no distribution for the average of dependent standardised statistics. Its grid-level e-value is 1.07, with a largest cell value of 1.29. The sign-timing results this study once reported at $p \approx 0.001$ leave no trace under an instrument that is valid for the design that produced them.

Read alongside the Clark–West columns, the table is not a contradiction but a scale statement. The implied information ratios—read off the realised growth rate through Theorem 7, $\mathrm{IR} = \sqrt{2P\Lambda}$—run from 0 to 0.51. The smallest ratio this sample could certify is 1.59 (1.11 with the stake pre-committed). The settings Clark–West likes are the ones whose implied ratios sit closest to that floor without reaching it. There is no disagreement between the instruments about the data; there is a disagreement about whether six years is enough to settle the question, and Corollary 9 says the certificate is right about that.

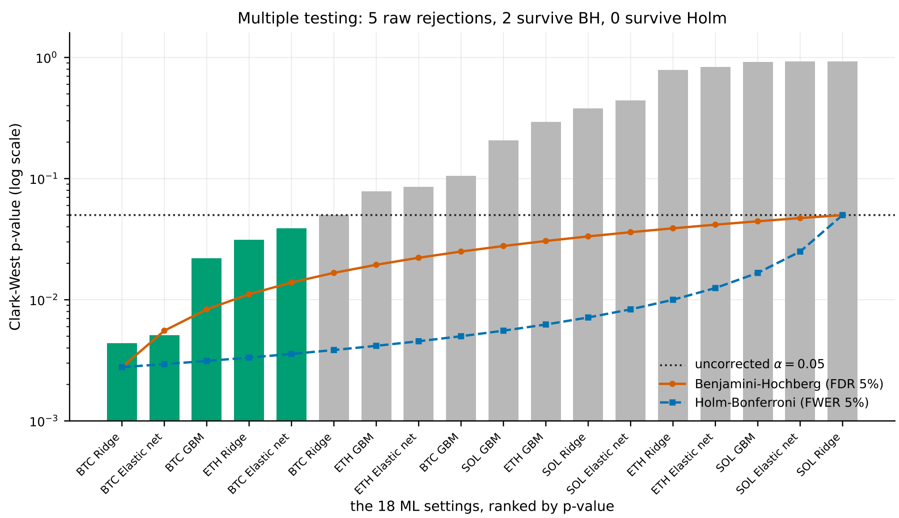

Figure 5: The 18 Clark–West p-values ranked against the Holm–Bonferroni and Benjamini–Hochberg thresholds, on a logarithmic scale because the entire decision occurs below $p = 0.05$. A bar must fall below a line to be rejected by that procedure. The figure's reference line is the nominal one; the calibrated expectation against this benchmark is roughly 4 rejections rather than 0.9 (Table 3).

8.4 What the features could still be worth

Failing to certify is the less useful half of an answer. The half a desk can act on is the anytime-valid ceiling of Section 7: an upper confidence bound, uniform over time, on the incremental value $\theta = \mathbb{E}[z_t(y_t - \mu)]$ of the standardised signal, with the drift eliminated exactly as in the certificate.

Expressed as annualised information ratios, the 95% ceilings in Table 6 run from 0.63 to 2.55, and every interval contains zero. The correct summary of this study is therefore neither "the features predict" nor "the features do not predict" but a statement with a number in it: *after six years of daily data, the incremental annualised information ratio of twelve standard technical features on these three assets lies between* 0.6 *and* 2.6 *depending on the setting, and cannot be distinguished from zero*. That is a bound a research manager can act on—it does not clear a costed hurdle for a strategy that must also carry market exposure—and it is a bound no p-value in this literature reports.

8.5 Multiple testing

Against the zero forecast, controlling the false discovery rate at 5% leaves BTC ridge and BTC elastic net at $h = 1$, both with $p^{\mathrm{BH}} = 0.046$, marginally under the line; controlling the family-wise error rate leaves nothing, because the smallest raw p-value, 0.0044, does not clear the Holm threshold of $0.05/18 = 0.0028$ (Figure 5). Those corrections are applied to p-values from a test whose size against this benchmark is 14–30% rather than 5%, so they do not control the error rates they name. This is not a technicality that cuts one way: the oversized test inflates the raw p-values' evidential reading, while the multiplicity adjustment is calibrated to a nominal level the test does not have. No calibrated bound on the false-discovery rate over this family can be extracted from the zero-benchmark column.

Against the recursive mean the adjustments are meaningful, because there the test is approximately correctly sized. The same two settings survive false-discovery control ($p^{\mathrm{BH}} = 0.035$), none survives family-wise control, and at the plug-in bandwidth the survivor counts are 5 and 1.

The count needs a joint null, not an expected value. An earlier version compared the observed number of rejections with the *sum* of the marginal rejection probabilities. That is $\mathbb{E}[N]$, not the distribution of N, and the eighteen tests are strongly dependent: ridge and elastic net are nearly the same estimator, the horizons share observations, the assets are cross-sectionally correlated, and every setting sees the same regimes. Under positive dependence false rejections cluster, so $P(N \geq 7)$ can

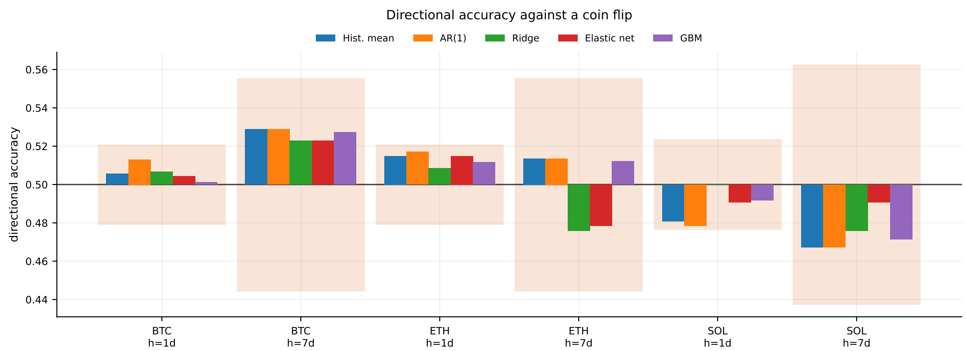

Figure 6: Directional accuracy against the band a fair coin would occupy. The band uses the effective sample size n/h: consecutive h-day forecasts are built from overlapping windows, and treating each as independent would shrink the band by $\sqrt{h}$ and manufacture significance.

be far larger than a binomial calculation with the same mean suggests. We therefore run the *entire* eighteen-setting experiment inside each null replication—three synthetic assets sharing a common sign process, two horizons, three estimators, all eighteen statistics—and read the joint quantities off the resulting vectors (`audit/scripts/mc_joint_null.py`). Across 2,000 replications the count has mean 0.96 and standard deviation 1.43, against 0.93 for eighteen independent tests: the dependence the reviewer of an earlier draft anticipated is real and inflates the spread by more than half. It does not rescue the count. The observed seven sits at $P(N \geq 7) = 0.0035$, with $P(N \geq 5) = 0.039$ and $P(N \geq 6) = 0.014$, and the marginal cell sizes run from 3.5% to 7.5% with a median of 4.9%, confirming the calibration of Table 3 setting by setting. Individually, however, nothing survives: the largest statistic, 2.667, sits just under the null's 95th percentile of 2.689, so the max-T global-null p-value is 0.052 and the smallest Romano–Wolf step-down adjusted p-value [45] is also 0.052. The honest reading is therefore split: the *number* of rejections is surprising under the joint null, and *no individual setting* is.

That is the strongest form the p-value evidence takes, and it is why Section 8.3 is not a contradiction of it: the certificate agrees the implied effect sits near the edge of what six years can resolve, and declines to call it.

8.6 Sign timing

Directional accuracy spans 0.471 to 0.528 and every setting lies inside its coin-flip band (Figure 6). No setting exhibits significant sign-timing skill.

The Pesaran–Timmermann variance terms all divide by n, so the statistic requires non-overlapping observations. An earlier version of this study fed it all n rows of an h-step forecast whose labels overlap $h-1$ times, which inflates the statistic by roughly $\sqrt{h}$. It was the same error Figure 6 avoids by drawing its band at n/h, so the table and the figure disagreed by a factor of $\sqrt{7}$ at the longer horizon. The pipeline now slices to $y_{k::h}$ for each phase offset k and averages the statistic over all h of them, and Table 7 reports that version throughout. The four largest $|S|$ at $h = 7$, before and after:

Setting	S (all n)	p	S (non-overlapping)	p
ETH elastic net, $h = 7$	-3.33	0.001	-1.28	0.242
ETH ridge, $h = 7$	-3.29	0.001	-1.28	0.265
SOL ridge, $h = 7$	-1.73	0.083	-0.67	0.604
BTC ridge, $h = 7$	$+1.41$	0.159	$+0.54$	0.596

The two ETH settings were previously reported as significantly *anti*-predictive at $p \approx 0.001$. They are not significant at any conventional level once the labels are non-overlapping, and neither is anything else. The conclusion—no sign-timing skill in either direction—is unchanged; the evidence previously offered for its most striking part was invalid.

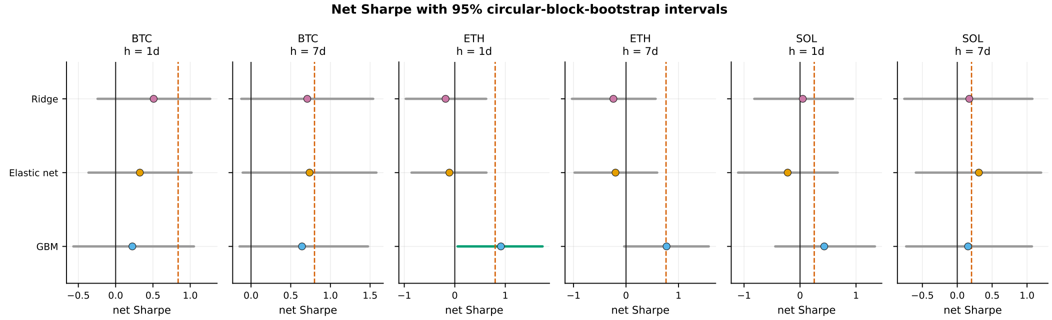

Figure 7: Net Sharpe with 95% circular-block-bootstrap intervals. The dashed line marks buy-and-hold on the same schedule and cost model. Intervals excluding zero are highlighted.

8.7 Economic performance

Two of 18 settings have a net Sharpe whose interval excludes zero, both gradient boosting on ETH: 0.91 with interval $[0.00, 1.74]$ at $h = 1$, and 0.92 with $[0.02, 1.80]$ at $h = 7$. Both intervals begin essentially at zero. Neither setting appears among the two surviving Benjamini–Hochberg correction, and neither of those two has an interval excluding zero.

The statistical winners and the economic winners are therefore disjoint. Two unrelated groups of survivors drawn from one search is the signature of noise: if a genuine short-horizon effect existed and were tradeable, the settings that detect it and the settings that profit from it ought to overlap.

Both economic winners are market exposure. Regressing each strategy's net return on buy-and-hold's, on the same schedule and cost model, with a Newey–West standard error on the intercept:

Setting	Sharpe	% long	β to buy-and-hold	α (ann.)	$t(\alpha)$
ETH GBM, $h = 1$	0.91	94%	0.76	+0.236	1.07
ETH GBM, $h = 7$	0.92	91%	0.89	+0.192	1.15
BTC GBM, $h = 7$	0.82	90%	0.82	+0.097	0.66
SOL elastic net, $h = 7$	0.60	56%	0.43	+0.499	1.23
Historical mean (all)	0.23–0.84	100%	1.00	≈ 0	$\leq \lvert 1.6 \rvert$

Neither of the two settings whose Sharpe interval excludes zero has an alpha distinguishable from zero once its market exposure is removed. An earlier version of this paper asserted the same thing in words, that the high-Sharpe results were long-only exposure; the regression is what makes it checkable.

Execution is assumed at a price the signal has not yet been computed from. Every feature at bar t uses the completed close C_t, and the backtest enters at that same C_t. Delaying entry by one bar, holding the forecasts fixed, moves the $h = 1$ results substantially and the $h = 7$ results hardly at all: ETH GBM at $h = 1$ falls from 0.91 to 0.21, BTC ridge at $h = 1$ from 0.51 to -0.23, ETH elastic net at $h = 1$ from -0.11 to -0.61; no $h = 7$ setting moves by more than 0.20. The one-day results are therefore properties of the execution convention as much as of the forecasts, and the surviving economic result—ETH GBM at $h = 7$, $0.92 \to 0.98$—is the one already shown above to be 91% long with $\beta = 0.89$.

One cost scenario is not a cost model. The 17 bp per side above is a single assumption. Sweeping it, with the forecasts held fixed, gives each setting a break-even cost—the per-side charge at which its net Sharpe reaches zero:

Per-side cost	0 bp	17 bp	25 bp	40 bp	100 bp
Settings with net Sharpe > 0	16/18	13/18	11/18	9/18	6/18

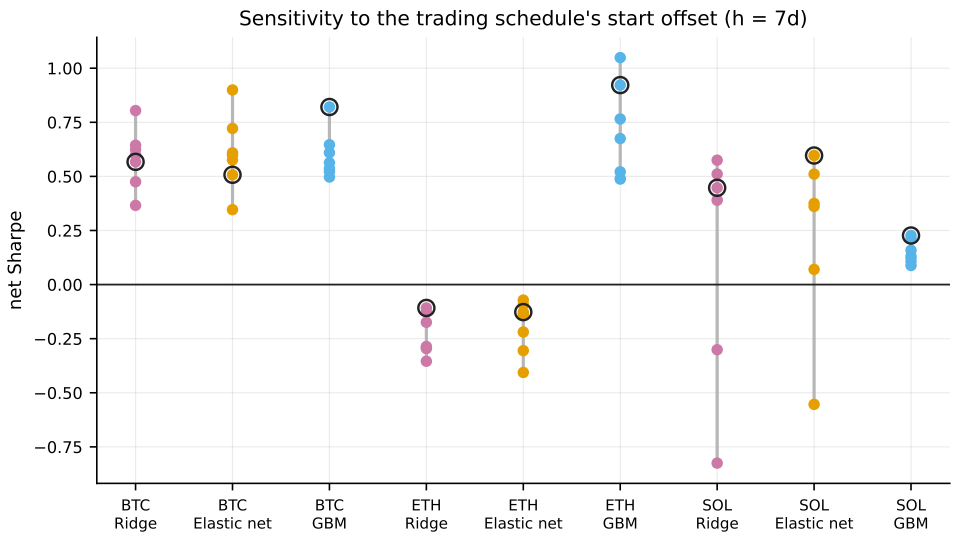

Figure 8: Net Sharpe on each of the h possible start offsets of the trading schedule at $h = 7$. The circled point is the offset the results table reports; nothing privileges it.

The median break-even cost is 31 bp, roughly twice the assumed charge, so the sign of the economic result is not delicate at the margin. What the sweep exposes instead is the exposure finding again, from a different direction. The settings that survive the highest costs are the ones that barely trade: ETH GBM at $h = 1$ breaks even at 148 bp and is 94% long, and the three settings with no finite break-even below 200 bp are all near-permanently long. Turnover, not forecast quality, is what determines cost tolerance here, and a strategy whose cost tolerance comes from never changing its mind is buy-and-hold with extra steps.

Six of 18 settings out-Sharpe buy-and-hold on the point estimate, the weakest form of that claim given that every interval contains zero. Meanwhile the historical-mean forecaster attains Sharpe ratios of 0.84, 0.80, and 0.25 across the three assets while making exactly one position change. It predicts a positive drift, hence is always long, and its net returns are numerically identical to buy-and-hold asset by asset and horizon by horizon. We carry buy-and-hold as an explicit row in Table 8 so that this cannot be mistaken for model performance.

Maximum drawdowns range from -55% to -99% across the machine-learning settings. This is a property of the position sizing rather than of any model: unit leverage reversing direction on roughly 70% annualised volatility carries a large variance drag, so the geometric return sits well below the arithmetic one. Buy-and-hold itself draws down between 76% and 96% over the same window.

8.8 Robustness to the trading schedule

Sampling every h-th bar leaves h schedules and nothing privileges the one starting at the first bar. At $h = 7$ the spread is substantial: SOL ridge reports a Sharpe of 0.45 drawn from a distribution running from -0.83 to 0.57, and SOL elastic net reports 0.60 from a range of $[-0.55, 0.60]$. A single-offset Sharpe at a multi-day horizon is therefore one draw from a distribution 1.4 Sharpe points wide, and the reported figure is in part a property of where sampling began.

9 Discussion

Estimating a nuisance and eliminating it are different operations. Diebold–Mariano is the default in applied forecast comparison and it is the wrong default whenever the benchmark is nested. Swapping in Clark–West fixes the nesting and leaves the hypothesis alone. Both benchmarks then dispose of the drift by plugging an estimate into the statistic, and the estimate's error does not disappear: with the zero forecast it is the whole drift and Proposition 1 prices it; with the recursive mean it is the covariance between the benchmark's estimation error and the model's signal, which is what the residual size distortion at $h = 7$ consists of. The certificate is the same comparison with the nuisance hedged rather than estimated, and the price of that—exactly $\ln 2$ nats, 23% more data—is small next to what it buys. The operational form of the point is a question a referee can ask in one line: does the larger model fit an intercept, and if so, against which of the two nulls is the reported p-value computed?

A distributional fix does not repair a hypothesis problem. Wild Clark–West [5] is the right tool for the problem it was built for—the core statistic is degenerate under the null, which is why its limit is non-standard—and it is well sized at long horizons in the authors' own designs. It cannot help here, and Section 4.9 shows that it does not: the drift term in Equation (12) survives the θ_t perturbation unchanged, because $\mathbb{E}[\theta_t] = 1$. Choosing the benchmark and choosing the reference distribution are separate decisions, and only the first determines what is being tested.

Validity that is proved does not have to be calibrated. The size distortion above is invisible in code review, survives a unit-test suite that checks every formula against hand-computed values, and is not detected by simulating a forecast that is independent noise. What detects it is a null built by resampling the data through the actual pipeline with actual estimated forecasts—and that experiment has to be redone whenever the feature set, the estimator, the sample or the horizon changes, at $2{,}000$ pipeline refits a time. Theorem 2 removes the obligation rather than automating it. The practical consequence is not only speed: a calibration study can only certify the pipeline it was run on, whereas a finite-sample theorem covers the pipeline a practitioner will build next year out of components that do not exist yet.

Most apparent performance was market exposure, and the certificate cannot be fooled by it. The highest Sharpe ratios in the study belong to a forecaster that makes one position change, and the strongest settings carry betas of 0.76 and 0.89 to buy-and-hold with alpha t-statistics near unity. Under Theorem 4 that exposure earns no evidence at all, because the certificate's bettor is credited with no part of the drift by construction. The exposure regression, the delayed-execution variant and the staggered portfolio remain worth running—they say *how* a strategy made its money—but they are no longer load-bearing for the inferential claim.

What is left, stated at the right scale. Against the correct benchmark and with size calibrated under a dependence-preserving null, Clark–West rejects in seven of eighteen settings against about one expected, and a joint resampling of the whole experiment puts that count in the tail. Under certificates nothing is certified, the grid-level e-value is 1.18 against 20, and the implied information ratios run from 0 to 0.50 against a certifiable floor of 1.59. These are not contradictory readings of the data; they are a p-value and an e-value disagreeing about what six years can settle, and Corollary 9 says the e-value is right. The honest summary is that the question is open at this sample size, and the number that makes it actionable is the ceiling: the features' incremental information ratio is bounded above by 2.6 and cannot be distinguished from zero. A desk can decide against that. It cannot decide against a p-value of 0.004 on a statistic whose null was never the one being asked about.

10 Limitations

Each item below is stated with what it would change if it were repaired, because a limitation that does not say that is a hedge.

Limitations that bound the reported results. The size calibration of Section 4.9 resamples all three assets' return series, so size is established for those return distributions and not for other frequencies or estimators outside the three used here; the booster is calibrated at $h = 1$ only. The sign-flipped null removes conditional mean predictability while preserving the volatility path, which is the dependence that matters most for daily returns, but it does not reproduce every feature of the real series—skewness dynamics and leverage effects, for instance, are destroyed with the sign. The minimum detectable effect is computed but large, so little here bounds predictability from above: a genuine R^2_{OS} of, say, $+0.002$ would be entirely consistent with what we observe: the minimum detectable effect measured in Section 4.9 is a population R^2_{OS} of 1.34% at 80% power, so this design is blind to anything an order of magnitude smaller. The multiplicity family of 18 covers one statistic over one grid, and the hyperparameters were fixed by hand and never tuned, so every conclusion is about $\mathrm{Ridge}(\alpha{=}1)$, $\mathrm{ElasticNet}(10^{-3}, 0.5)$, and one booster configuration rather than about a model family; a tuned search would need its own multiplicity accounting, which would raise the bar rather than lower it. The sample end date was unpinned until this revision—the default configuration resolved it to the current day, so any run missing the committed cache silently re-dated the study—and the consequence was not hypothetical: one Monte Carlo pass was measured on a sample twelve bars longer than the manuscript's and reported cell values differing by up to five percentage points. `StudyConfig.end` is now pinned to 2026-07-18 in the configuration rather than only by the presence of the cache.

Limitations of the certificate. Three are worth naming and none is hidden by the theorem. *The assumption is not free.* The default payoff needs the outcome to be conditionally symmetric about its drift; that permits arbitrary volatility clustering and tail thickness but excludes systematically skewed conditional distributions, and daily cryptocurrency returns are mildly skewed. The identity payoff drops the assumption to a martingale difference at a cost in power set by how loose the a-priori return envelope is. Run on this study with an envelope of $\sqrt{h}$, it gives a largest e-value of 4.13 and a grid-level e-value of 1.44, against 1.81 and 1.18 for $\tanh$: larger, still an order of magnitude short of the 20 required, and it selects nothing under e-BH. The conclusions of Table 6 do not depend on the symmetry assumption. *The power is real but bounded.* Corollary 9 says no anytime-valid test does better in order, but a fixed-sample test that is pre-committed, looked at once, and handed the true drift needs 23% less data; a study that genuinely will not look at its data until a pre-registered sample size is reached gives up nothing by using one. *The infimum is numerical.* The drift search is a grid, and a grid too coarse to resolve $\sigma/\sqrt{n}$ silently destroys the guarantee rather than merely weakening it. We report the refinement check, the library defaults conservatively and exposes the spacing, and its test suite records the failure mode; that is mitigation, not elimination.

Limitations that bound generalisation. Asset selection is not survivorship-free (Section 3); this biases towards finding profitable long exposure, which strengthens a negative result about predictability and would invalidate any cross-sectional claim. We use daily spot bars only, with no order book, funding rates, intraday structure, or cross-exchange information, and short-horizon predictability, if it exists, is more likely to live at frequencies this data cannot resolve—so the negative result should be read as being about daily bars specifically. Short positions are modelled as fully collateralised, returning $-r$ with no borrow cost, margin, or liquidation, which is optimistic; correcting it would lower the reported Sharpe ratios and cannot raise them. Costs are a flat 17 bp per side that does not scale with size or volatility; we report a break-even sweep in Section 8.7 but not a queue-level execution model. Intervals are percentile bootstrap rather than BCa, adequate for a reject-or-not reading and mildly biased for a statistic as asymmetric as the Sharpe ratio. The deflated Sharpe ratio deflates for the six models raced within one asset–horizon—three of which are benchmarks—rather than for the whole grid or for the choices fixed before the grid was run, so it is an upper bound on the true corrected figure. Tightening it would not change a conclusion: the largest deflated Sharpe among the 18 settings is 0.847 (BTC GBM, $h = 7$), already short of any conventional threshold.

11 Conclusion

Tests of out-of-sample predictability dispose of the asset's drift by estimating it, and that is where they break. Proposition 1 makes the failure exact: against a zero benchmark the Clark–West statistic converges to the t-statistic of the asset's mean return, so on an asset with Bitcoin's drift a model containing no features at all is declared significantly predictive in 43% of samples, and with twelve estimated coefficients it is still 12%. Replacing zero with the recursive mean removes the leading term and leaves the benchmark's own estimation error behind.

The alternative developed here does not estimate the drift. A certificate is a non-negative wealth process—one bettor staking on the signal, one hedging every candidate drift—whose infimum over the nuisance is bounded by a martingale under the entire composite null. Validity is finite-sample and holds at every stopping time, with no bandwidth, no long-run variance, no bootstrap, no asymptotics and no refitting, for arbitrary black-box pipelines. Three things follow that a t-statistic cannot supply: the process may be monitored daily on a live strategy without any correction for looking; its logarithm is the profit and loss of an explicit drift-hedged strategy, so a 5% rejection *is* a twentyfold multiplication of capital that was credited with none of the market's return; and because e-values average under arbitrary dependence, an eighteen-setting grid needs no joint bootstrap and an overlapping h-step forecast is combined by splitting capital across its h daily vintages—the staggered portfolio, arrived at from the statistics rather than bolted on afterwards.

The construction also closes the sample-size question. Certifying an annualised information ratio IR takes $2\ln(1/\alpha)/\mathrm{IR}^2$ years, about $6/\mathrm{IR}^2$ at the 5% level, rising to $7.4/\mathrm{IR}^2$ once the drift hedge is paid for and to $16/\mathrm{IR}^2$ if the stake is learned online rather than declared in advance. The bound is optimal to first order, so no better test recovers the missing years, and continuous monitoring costs only three per cent more data than a fixed-sample test that may be looked at once. Six years of daily data—about what any liquid cryptocurrency offers—cannot certify an information ratio below 1.59.

Applied to this study, nothing is certified. The grid-level e-value is 1.18 against the 20 required, the directional variant that replaces an invalid market-timing test gives 1.07, and the implied information ratios run from 0 to 0.50. We do not read that as evidence of no predictability, and the instrument says why: the settings Clark–West likes imply ratios just below what six years can resolve. The reportable number is therefore not a p-value but a ceiling—after six years, the incremental information ratio of twelve standard technical features on these three assets is at most 2.6, and cannot be distinguished from zero. That is a statement a research manager can act on, and it is available at any time, on any pipeline, for the cost of one pass over forecasts that have already been computed.

Reproducibility. All results, figures, and tables in this paper are generated by a single command from a public repository, and the tables reproduced here are emitted directly from the committed results file rather than transcribed. The test suite covers the leakage guarantees and checks the statistical formulas against hand-computed reference values. It did not, and could not, detect the benchmark misspecification of Section 4.8: every formula was implemented correctly and every test passed. The size calibration of Section 4.9, the corrected-benchmark re-test, the Wild Clark–West implementation, the non-overlapping sign tests, the exposure regressions, and the delayed-execution backtest are separate scripts under `audit/`, each reproducing the numbers quoted for it, together with a full record of which earlier claims they overturn. The per-observation forecasts on which every test in this paper is computed are committed as a file, so the inference can be re-analysed without re-running the models. Two cautions apply to anyone repeating the size experiment. It must be run with the sample end date pinned, for the reason given in Section 10; and it needs a serious replication count, because at 400 replications we observed cell-to-cell movement of up to three percentage points between random seeds. The certificate needs neither caution, because it needs no size experiment; the one thing it does need is a drift grid fine enough to resolve the mean, and the library's default and its test suite both enforce that.

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A Proofs and derivations

Theorem 7, in full. Let $\psi_t = g_t\phi_t(\mu_0)$ with $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \eta_t$ under the alternative. For a stake λ small enough that $\lambda|\psi| \le c < 1$,

$$\mathbb{E}\log(1+\lambda\psi) = \lambda\,\mathbb{E}[\psi] - \tfrac{\lambda^2}{2}\mathbb{E}[\psi^2] + O(\lambda^3),$$

maximised at $\lambda^* = \mathbb{E}[\psi]/\mathbb{E}[\psi^2]$ with value $\Lambda = \mathbb{E}[\psi]^2/(2\mathbb{E}[\psi^2]) + O(\mathbb{E}[\psi]^3)$. Take ϕ the identity, $s_t \equiv \sigma$, and $g_t \propto m_t$ where $m_t = \mathbb{E}[y_t \mid \mathcal{F}_{t-1}] - \mu_0$ is the conditional mean. Then $\mathbb{E}[\psi] \propto \mathbb{E}[m^2]/\sigma$ and $\mathbb{E}[\psi^2] \propto \mathbb{E}[m^2]$ to leading order in $\mathbb{E}[m^2]/\sigma^2$, so

$$\Lambda \;=\; \frac{\mathbb{E}[m^2]}{2\sigma^2}\big(1+o(1)\big) \;=\; \frac{\mathrm{IR}_p^2}{2}\big(1+o(1)\big),$$

because the strategy holding m_t/σ units earns $\mathbb{E}[m^2]/\sigma$ per period with standard deviation $\sqrt{\mathbb{E}[m^2]}$, hence per-period information ratio $\mathrm{IR}_p = \sqrt{\mathbb{E}[m^2]}/\sigma$. Crossing $\log(1/\alpha)$ therefore takes $T^* =$

$2\log(1/\alpha)/\mathrm{IR}_p^2$ periods. For a bounded odd payoff the same calculation runs with ϕ in place of the identity and multiplies Λ by $\mathbb{E}[\phi'(\varepsilon/\sigma)]^2/\mathbb{E}[\phi(\varepsilon/\sigma)^2]$, which for $\phi = \tanh$ and Gaussian ε is 0.98: the bounded transform costs about two per cent of the growth rate and buys the removal of the envelope, which for daily returns is worth a factor of two in the other direction.

Corollary 9. Under the alternative with conditional mean shift m_t and Gaussian innovations, the per-period Kullback–Leibler divergence from H_0 is $\mathbb{E}[m^2]/(2\sigma^2) = \mathrm{IR}_p^2/2$. Wald's identity for a level-α sequential test with vanishing type-II error gives $\mathbb{E}[\tau] \geq \log(1/\alpha)/\mathrm{KL}$, which is T^*. Hence the constant in Theorem 7 is not improvable by a different anytime-valid construction.

Corollary 11. With a stake grid $\{\lambda_1, \ldots, \lambda_G\}$ and uniform prior, $\frac{1}{G}\sum_j \prod_t(1+\lambda_j\psi_t) \geq \frac{1}{G}\max_j \prod_t(1+\lambda_j\psi_t)$, so log wealth is within $\log G$ of the best fixed stake in hindsight; a continuous mixture in the style of Cover [32] attains $\frac{1}{2}\log T + O(1)$, and no online rule does better in order. Substituting into $T\mathrm{IR}_p^2/2 = \log(1/\alpha) + \rho_T$ and solving gives the "learned online" column of Table 4.

Why the recursive mean does not finish the job. With $b_t = \hat{\mu}_t$ the Clark–West differential is $\hat{f}_t = 2(y_t - \hat{\mu}_t)g_t$ with $g_t = \hat{y}_t^m - \hat{\mu}_t$, so

$$\mathbb{E}[\hat{f}_t] = 2\,\mathbb{E}\big[(y_t - \mu)g_t\big] - 2\,\mathbb{E}\big[(\hat{\mu}_t - \mu)g_t\big].$$

The first term is the estimand. The second is the covariance between the benchmark's own estimation error and the model's signal, both functions of the same training window; it does not vanish in finite samples, it grows with the horizon because overlapping labels make $\hat{\mu}_t$ and g_t share more observations, and it is what the residual 7–8% size at $h = 7$ in Table 3 consists of. Equation (18) removes it because no estimate of μ appears anywhere in the construction.

B Audit trail of corrected claims

Earlier versions of this study reported findings that did not survive their own robustness checks. They are listed here rather than in the body, because the scientific content is in the current sections and a reader who wants the forensic record should be able to find it in one place. The full changelog, with the script that produced each number, is in the repository.

Benchmark. "Five of 18 settings reject, against 0.9 expected from a search of that size." The comparison assumed the test was correctly sized for the hypothesis intended. It was not: against the zero forecast the measured rejection rate on data with no predictability is 14–21% at $h = 1$, so the correct expectation was about four. Proposition 1 now gives the distortion in closed form.

The nesting claim. The manuscript asserted that the zero forecast is the slopes-zero restriction of the estimators. It is not; the restriction is the training-window mean, because the intercept is unpenalised.

The validating simulation. An earlier calibration replaced the larger model's forecast with exogenous noise. That construction makes the null of equal accuracy false rather than true and estimates no coefficient, so it measures neither size nor bias. It was replaced by refitting the entire pipeline inside each replication.

Sign timing. Significant market-timing results at $p \approx 0.001$ came from treating overlapping seven-day labels as independent. On non-overlapping subsamples they are $p \approx 0.24$–0.27. The phase-averaged repair introduced at that point was itself invalid—it referred an average of dependent standardised statistics to a standard normal without deriving a variance for the average—and is superseded by the directional certificate of Section 8.3.

Execution and schedule. The primary backtest entered at the same close its features were computed from, which is infeasible, and reported one of h arbitrary starting phases. Both are replaced: execution is delayed by one bar and the strategy is the staggered portfolio over all h vintages, which Remark 6 shows is also the statistically correct combination.

Size measurement. The first Monte Carlo used 400 replications against a sample that a working-directory bug had silently re-downloaded twelve bars longer. Cells moved by up to three percentage points between seeds. Rebuilt at 2,000 replications against the pinned cache.

The dependence null. A 28–43% rejection rate under block resampling was read as size failure. Block resampling preserves whatever genuine within-block predictability the real series contains, so that rate is an upper bound on size, not a measurement of it. The sign-flipped null separates the two.

Novelty. An earlier version presented the benchmark correction as its main contribution. A literature check established that reporting both benchmarks is standard practice in this literature [6, 4, 2, 3]. The contribution was narrowed accordingly, and the present version claims novelty only for the instrument of Sections 5–6.

Each of these was found by a check the study already ran and then reported honestly, which is the only reason they are known. The pattern worth extracting is that none of them was visible in the code: every one was a mismatch between a quantity the code computed correctly and the sentence it was used to support.

C Statistical definitions

Pesaran–Timmermann. With P the hit rate, P_y and P_x the proportions of positive outcomes and forecasts, and $P_* = P_y P_x + (1-P_y)(1-P_x)$,

$$S = \frac{P - P_*}{\sqrt{\widehat{\text{var}}(P) - \widehat{\text{var}}(P_*)}} \xrightarrow{d} \mathcal{N}(0,1), \qquad \widehat{\text{var}}(P) = \frac{P_*(1-P_*)}{n}, \tag{19}$$

$$\widehat{\text{var}}(P_*) = \frac{(2P_y-1)^2 P_x(1-P_x)}{n} + \frac{(2P_x-1)^2 P_y(1-P_y)}{n} + \frac{4P_yP_x(1-P_y)(1-P_x)}{n^2}. \tag{20}$$

Every term divides by n, so n must count independent observations. Section 8.6 reports the statistic on non-overlapping subsamples for that reason.

Probabilistic and deflated Sharpe ratio. With skewness γ_3 and kurtosis γ_4,

$$\widehat{\text{PSR}}(\text{SR}^*) = \Phi\left[\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3\widehat{\text{SR}} + \frac{\gamma_4 - 1}{4}\widehat{\text{SR}}^2}}\right], \tag{21}$$

and the deflated Sharpe ratio is $\widehat{\text{PSR}}$ evaluated at $\text{SR}^* = \mathbb{E}[\max_{i\le N} \text{SR}_i]$, approximated by

$$\mathbb{E}\left[\max_{i\le N} \text{SR}_i\right] \approx \sigma_{\text{SR}}\left[(1-\gamma)\,\Phi^{-1}\big(1 - \tfrac{1}{N}\big) + \gamma\,\Phi^{-1}\big(1 - \tfrac{1}{Ne}\big)\right], \tag{22}$$

with γ the Euler–Mascheroni constant.

Wild Clark–West. Let $\hat{e}_t^{\,b} = y_t - \hat{y}_t^{\,b}$ and $\hat{e}_t^{\,m} = y_t - \hat{y}_t^{\,m}$. Pincheira et al. [5] replace the Clark–West core statistic $\hat{f}_t = 2\hat{e}_t^{\,b}(\hat{e}_t^{\,b} - \hat{e}_t^{\,m})$ with

$$\hat{f}_t^{\,\text{W}} = \hat{e}_t^{\,b}\big(\hat{e}_t^{\,b} - \theta_t\,\hat{e}_t^{\,m}\big), \qquad \theta_t \overset{\text{iid}}{\sim} \mathcal{N}(1, \phi^2), \tag{23}$$

with θ_t independent of the data. Under the null the two models coincide, so $\mathbb{E}[\hat{f}_t^{\,\text{W}}] = \mathbb{E}[(\hat{e}_t^{\,b})^2(1-\theta_t)] = 0$ while the variance stays strictly positive, which restores the rank condition of West [18] and with it asymptotic normality. We set $\phi = c\,\widehat{\text{sd}}(\hat{e}^{\,m})$ with $c \in \{0.01, 0.02, 0.04\}$ and average $K = 2$ independent draws as the authors prescribe, dividing by $\sqrt{\sum_i \sum_j \rho_{ij}}$ where ρ_{ij} is the sample correlation between the ith and jth core sequences.

D Full results

Table 7 gives forecast accuracy and the predictive-accuracy tests for every model, asset, and horizon. Clark–West appears in two columns, against the zero forecast and against the recursively estimated mean; per Section 4.8 these are different hypotheses rather than a robustness check, and the second is the one whose size is calibrated in Table 3. Table 8 gives net-of-cost performance with bootstrap intervals, the phase spread, and both Sharpe corrections. Entries in bold are significant at the uncorrected 5% level; Section 8.2 gives the corrected counts, which are what should be read.

Table 7: Forecast accuracy and predictive-accuracy tests, all settings. Each test is reported as statistic (p). DM negative favours the model, CW positive favours the model, PT positive indicates sign-timing skill.

Model	RMSE	R^2_{OS}	Dir. acc.	Rank IC	DM (p)	CW vs. 0 (p)	CW vs. mean (p)	PT (p)
BTC, $h = 1$								
Random walk	0.0301	0.0005	–	–	–	–	+1.38 (0.084)	–
Historical mean	0.0301	0.0000	0.506	0.002	+0.18 (0.860)	+1.03 (0.152)	–	–
AR(1)	0.0301	-0.0013	0.513	0.033	+0.34 (0.737)	+1.24 (0.107)	+0.86 (0.194)	+1.23 (0.217)
Ridge	0.0302	-0.0079	0.507	0.049	+0.76 (0.447)	**+2.62 (0.004)**	**+2.66 (0.004)**	+0.54 (0.591)
Elastic net	0.0301	0.0031	0.504	0.046	-0.46 (0.644)	**+2.57 (0.005)**	**+2.67 (0.004)**	+0.23 (0.817)
GBM	0.0301	0.0007	0.501	0.019	-0.04 (0.970)	**+2.02 (0.022)**	**+1.75 (0.040)**	-0.39 (0.698)
BTC, $h = 7$								
Random walk	0.0804	0.0072	–	–	–	–	**+1.80 (0.036)**	–
Historical mean	0.0807	0.0000	0.529	-0.018	+0.40 (0.686)	+1.01 (0.156)	–	–
AR(1)	0.0807	-0.0002	0.529	-0.023	+0.42 (0.675)	+1.00 (0.159)	-0.36 (0.639)	–
Ridge	0.0825	-0.0466	0.523	0.044	+1.85 (0.064)	+1.64 (0.050)	+1.36 (0.086)	+0.54 (0.596)
Elastic net	0.0817	-0.0258	0.523	0.043	+1.43 (0.152)	**+1.76 (0.039)**	+1.54 (0.061)	+0.47 (0.633)
GBM	0.0809	-0.0055	0.527	0.032	+0.57 (0.569)	+1.25 (0.105)	+1.02 (0.153)	+0.28 (0.723)
ETH, $h = 1$								
Random walk	0.0407	0.0006	–	–	–	–	+1.30 (0.097)	–
Historical mean	0.0408	0.0000	0.515	0.019	+0.25 (0.806)	+0.81 (0.210)	–	–
AR(1)	0.0408	-0.0020	0.517	0.036	+0.47 (0.639)	+1.16 (0.124)	+0.91 (0.181)	+0.95 (0.345)
Ridge	0.0411	-0.0167	0.508	0.053	+1.42 (0.157)	**+1.86 (0.031)**	**+1.76 (0.039)**	+0.52 (0.600)
Elastic net	0.0409	-0.0051	0.515	0.040	+0.82 (0.413)	+1.37 (0.085)	+1.25 (0.106)	+1.00 (0.315)
GBM	0.0408	-0.0009	0.512	0.021	+0.29 (0.775)	+1.42 (0.078)	+1.08 (0.139)	-0.30 (0.762)
ETH, $h = 7$								
Random walk	0.1073	0.0057	–	–	–	–	**+1.66 (0.048)**	–
Historical mean	0.1076	0.0000	0.514	0.039	+0.39 (0.696)	+0.88 (0.188)	–	–
AR(1)	0.1076	-0.0002	0.514	0.034	+0.41 (0.685)	+0.87 (0.191)	-0.31 (0.623)	–
Ridge	0.1124	-0.0908	0.476	-0.047	**+3.15 (0.002)**	-0.79 (0.786)	-1.63 (0.948)	-1.27 (0.265)
Elastic net	0.1115	-0.0749	0.479	-0.052	**+2.98 (0.003)**	-0.98 (0.837)	-1.80 (0.964)	-1.28 (0.242)
GBM	0.1083	-0.0126	0.512	-0.042	+1.19 (0.234)	+0.55 (0.292)	-0.36 (0.640)	+0.09 (0.403)
SOL, $h = 1$								
Random walk	0.0499	0.0153	–	–	–	–	**+4.75 (0.000)**	–
Historical mean	0.0502	0.0000	0.481	-0.029	**+3.00 (0.003)**	-1.23 (0.891)	–	–
AR(1)	0.0504	-0.0062	0.478	-0.007	**+2.39 (0.017)**	-0.77 (0.779)	-0.41 (0.658)	-0.81 (0.418)
Ridge	0.0509	-0.0262	0.500	0.010	**+2.10 (0.036)**	+0.31 (0.379)	+1.34 (0.091)	+0.01 (0.996)
Elastic net	0.0504	-0.0080	0.491	0.013	+1.56 (0.119)	+0.15 (0.441)	+1.43 (0.077)	-0.66 (0.506)
GBM	0.0505	-0.0095	0.492	0.023	**+2.00 (0.046)**	+0.82 (0.207)	**+1.87 (0.031)**	+0.57 (0.569)
SOL, $h = 7$								
Random walk	0.1346	0.1023	–	–	–	–	**+5.23 (0.000)**	–
Historical mean	0.1420	0.0000	0.467	-0.103	**+3.63 (0.000)**	-1.74 (0.959)	–	–
AR(1)	0.1420	0.0007	0.467	-0.099	**+3.62 (0.000)**	-1.71 (0.957)	+1.29 (0.099)	–
Ridge	0.1449	-0.0416	0.476	-0.039	**+3.22 (0.001)**	-1.43 (0.923)	**+2.00 (0.022)**	-0.67 (0.604)
Elastic net	0.1435	-0.0210	0.491	-0.034	**+3.19 (0.001)**	-1.41 (0.921)	**+2.28 (0.011)**	-0.19 (0.504)
GBM	0.1447	-0.0379	0.471	-0.081	**+3.32 (0.001)**	-1.39 (0.918)	+0.37 (0.355)	-0.18 (0.680)

Table 8: Net-of-cost performance. Sharpe ratios annualise at $365/h$. Phases gives the range of net Sharpe across all h start offsets of the trading schedule. DSR is an upper bound, as discussed in Section 10.

Model	Sharpe	95% CI	Phases	Max DD	PSR	DSR
BTC, $h = 1$						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.84	[0.01, 1.63]	–	-76.6%	0.98	0.87
AR(1)	0.21	[-0.52, 0.91]	–	-82.7%	0.70	0.34
Ridge	0.51	[-0.13, 1.25]	–	-65.7%	0.89	0.63
Elastic net	0.33	[-0.35, 1.00]	–	-74.3%	0.79	0.45
GBM	0.22	[-0.55, 1.05]	–	-89.4%	0.71	0.35
Buy & hold	0.84	[0.01, 1.63]	–	-76.6%	0.98	–
BTC, $h = 7$						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.80	[-0.12, 1.63]	[0.78, 0.80]	-75.9%	0.98	0.83
AR(1)	0.80	[-0.12, 1.63]	[0.78, 0.80]	-75.9%	0.98	0.83
Ridge	0.57	[-0.27, 1.44]	[0.37, 0.80]	-78.5%	0.92	0.65
Elastic net	0.51	[-0.54, 1.55]	[0.35, 0.90]	-86.1%	0.90	0.60
GBM	0.82	[-0.08, 1.69]	[0.50, 0.82]	-54.6%	0.98	0.85
Buy & hold	0.80	[-0.12, 1.63]	–	-75.9%	0.98	–
ETH, $h = 1$						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.80	[-0.04, 1.59]	–	-79.4%	0.97	0.64
AR(1)	-0.15	[-0.89, 0.60]	–	-93.7%	0.36	0.02
Ridge	-0.18	[-0.93, 0.61]	–	-97.5%	0.33	0.02
Elastic net	-0.11	[-0.86, 0.63]	–	-97.7%	0.39	0.03
GBM	0.91	[0.02, 1.74]	–	-80.4%	0.99	0.74
Buy & hold	0.80	[-0.04, 1.59]	–	-79.4%	0.97	–
ETH, $h = 7$						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.76	[-0.09, 1.56]	[0.71, 0.76]	-77.3%	0.97	0.62
AR(1)	0.76	[-0.09, 1.56]	[0.71, 0.76]	-77.3%	0.97	0.62
Ridge	-0.11	[-0.98, 0.81]	[-0.35, -0.11]	-99.2%	0.40	0.03
Elastic net	-0.13	[-0.95, 0.72]	[-0.41, -0.07]	-98.9%	0.38	0.03
GBM	0.92	[0.02, 1.80]	[0.49, 1.05]	-84.5%	0.99	0.76
Buy & hold	0.76	[-0.09, 1.56]	–	-77.3%	0.97	–
SOL, $h = 1$						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.25	[-0.68, 1.22]	–	-96.3%	0.71	0.43
AR(1)	-0.19	[-0.99, 0.69]	–	-97.5%	0.34	0.13
Ridge	0.05	[-0.85, 0.82]	–	-97.4%	0.54	0.27
Elastic net	-0.22	[-1.17, 0.60]	–	-97.1%	0.32	0.12
GBM	0.43	[-0.52, 1.25]	–	-91.3%	0.83	0.59
Buy & hold	0.25	[-0.68, 1.22]	–	-96.3%	0.71	–
SOL, $h = 7$						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.24	[-0.80, 1.19]	[0.20, 0.25]	-96.2%	0.70	0.47
AR(1)	0.24	[-0.80, 1.19]	[0.20, 0.25]	-96.2%	0.70	0.47
Ridge	0.45	[-0.43, 1.23]	[-0.83, 0.57]	-89.7%	0.84	0.65
Elastic net	0.60	[-0.23, 1.34]	[-0.55, 0.60]	-83.9%	0.91	0.77
GBM	0.23	[-0.69, 1.03]	[0.09, 0.23]	-94.9%	0.69	0.46
Buy & hold	0.24	[-0.80, 1.19]	–	-96.2%	0.70	–

E Equity curves

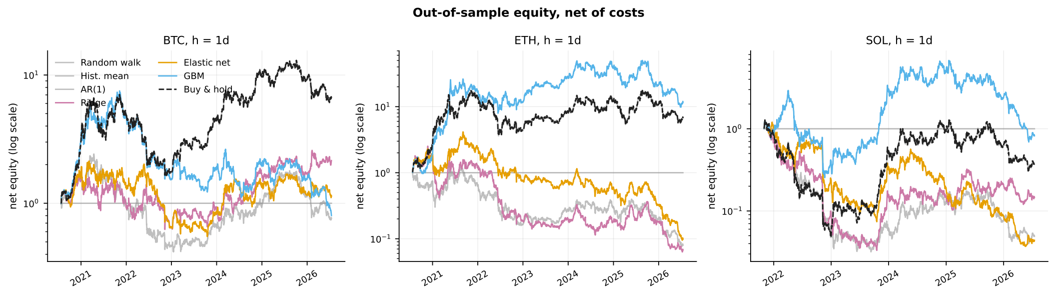

Figure 9: Out-of-sample equity net of costs at $h = 1$, against the always-long reference on the same schedule and cost model. Starting capital is unity and the vertical scale is logarithmic.